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OmniXtreme: Breaking the Generality Barrier in High-Dynamic Humanoid Control

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Project page: https://extreme-humanoid.github.io/

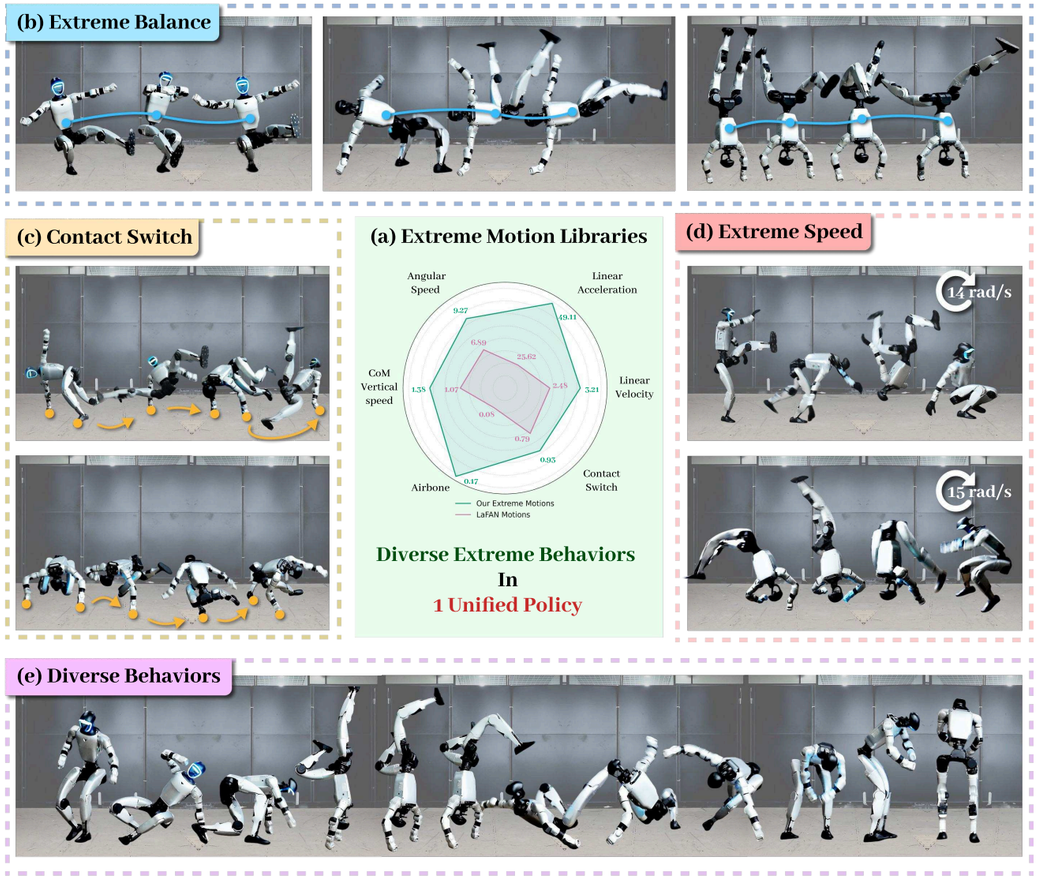


Fig. 1: **Extreme whole-body humanoid control from our unified policy OMNIXTREME.** (a) A quantitative comparison shows that our curated extreme motion libraries occupy substantially more challenging regimes than standard multi-motion benchmarks (*e.g.*, Unitree-retargeted LAFAN1). Real-world executions of our unified policy OMNIXTREME demonstrate robust and physically executable extreme behaviors drawn from this motion set, including (b) extreme balance behaviors, (c) rapid contact switching with complex support transitions, (d) high-speed motions with large angular velocities, and (e) diverse whole-body behaviors spanning qualitatively distinct motion styles.

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OmniXtreme：突破高动态人形机器人控制的泛化瓶颈

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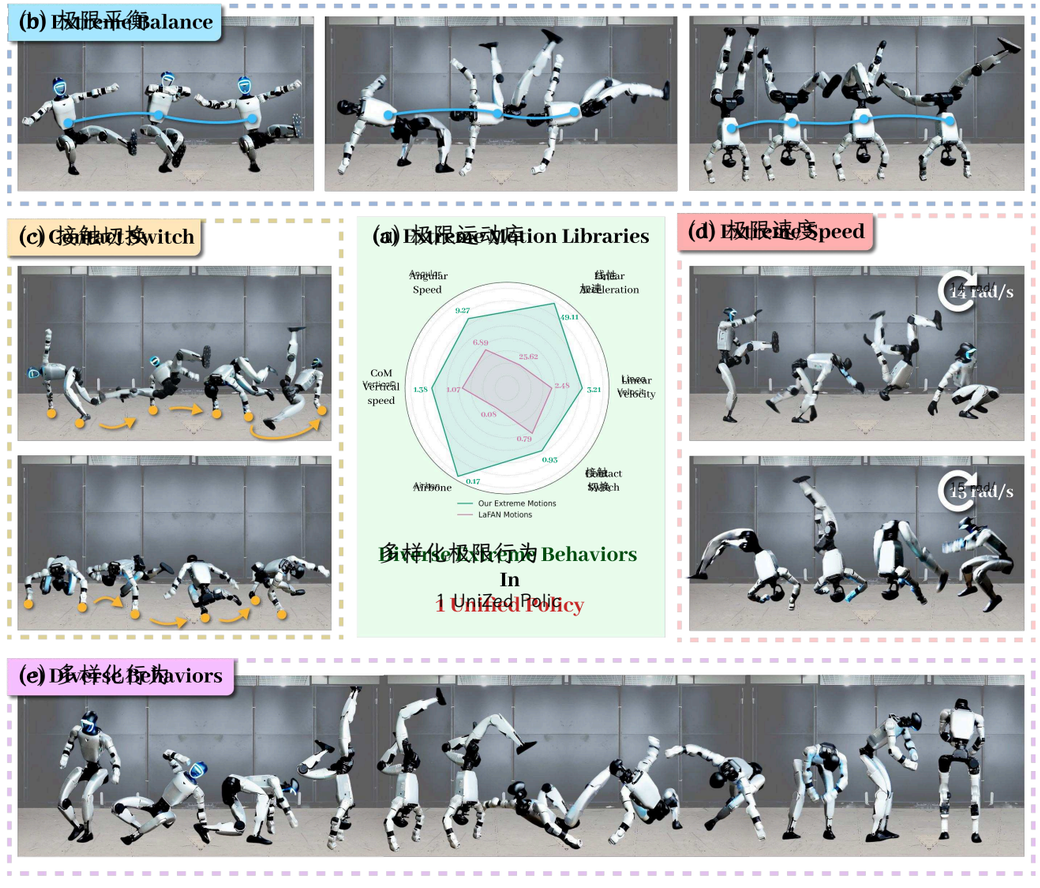


图1：我们统一的策略OMNIXTREME实现的极限全身人形控制。(*a*) 定量比较显示，我们精选的极限运动库占据的挑战性区域远超标准多运动基准（例如欢腾科技重定向的LAFAN1）。我们统一策略OMNIXTREME在现实世界中的执行展示了从该运动集提取的鲁棒且可物理执行的极限行为，包括 (b) 极限平衡行为，(c) 快速接触切换与复杂支撑转换，(d) 大角速度的高速运动，以及 (*e*) 跨越定性不同运动风格的多样化全身行为。

***Abstract*—High-fidelity motion tracking serves as the ultimate litmus test for generalizable, human-level motor skills. However, current policies often hit a "generality barrier": as motion libraries scale in diversity, tracking fidelity inevitably collapses—especially for real-world deployment of high-dynamic motions. We identify this failure as the result of two compounding factors: the learning bottleneck in scaling multi-motion optimization and the physical executability constraints that arise in real-world actuation. To overcome these, we introduce OMNIXTREME, a scalable framework that decouples general motor skill learning from sim-to-real physical skill refinement. Our approach uses a flow-matching policy with high-capacity architectures to scale representation capacity without the interference-intensive multi-motion RL optimization, followed by an actuation-aware refinement phase that ensures robust performance on physical hardware. Extensive experiments demonstrate that OMNIXTREME maintains high-fidelity tracking across diverse, high-difficulty datasets. On real robots, the unified policy successfully executes multiple extreme motions, effectively breaking the long-standing fidelity–scalability trade-off in high-dynamic humanoid control.**

I. INTRODUCTION

We ultimately seek general-purpose humanoids with scalable, human-level whole-body motor skills. A natural and widely used way to study such capability is high-fidelity motion tracking, where a controller must reproduce reference motions accurately while remaining dynamically stable under contacts and disturbances. High-quality tracking is more than an aesthetic goal: it captures whole-body coordination and contact timing that underlie loco-manipulation, expressive interaction, and many downstream core humanoid capabilities [59, 50, 52, 1, 43, 47, 53].

Over the past years, learning-based motion tracking has made striking progress: with carefully designed objectives and reinforcement learning, policies can track individual motions with high precision, including highly dynamic behaviors such as dance, flips, and martial arts [57, 28, 14]. More recent work [5, 30, 9, 23, 11, 6, 12, 58, 55, 26] has taken important steps toward multi-motion controllers that cover broader behavior libraries. Yet a recurring pattern persists: when we scale to larger, more heterogeneous motion libraries spanning diverse styles, contact regimes, and timing modes, motion tracking quality tends to degrade. Controllers become conservative and "average," break on the hardest motions, or prove brittle to the small deviations that inevitably occur in sim-to-real transfer. The degradation is particularly pronounced in *high-dynamic motions*, where even small tracking errors can rapidly cascade into catastrophic failures. This long-standing fidelity–scalability trade-off has effectively capped the level of generality achievable in humanoid motor control, particularly in *high-dynamic regimes*, suggesting a fundamental limitation rather than isolated engineering issues [11, 58, 14, 57].

A central question therefore arises: **why is high-fidelity motion tracking so difficult to scale, especially on real humanoid robots?** We argue that this difficulty stems from two compounding barriers that emerge at different stages of the current sim-to-real training pipeline.

The first barrier is the **learning bottleneck** that arises even in simulation. Several recent works [58, 5, 23, 30, 55, 26, 12, 11] have begun to explore multi-motion humanoid tracking, aiming to improve scalability beyond single-motion imitation. However, existing approaches remain constrained by both representation and optimization. On the representation side, most approaches rely on relatively simple policy parameterizations, such as MLP actors [58, 5, 23, 55, 26, 12, 11, 46]. When required to map observations to highly heterogeneous action targets arising from diverse behaviors and contact patterns, such parameterizations have been observed to exhibit limited scalability as data diversity increases [33]. On the optimization side, jointly training a unified policy across many motions using reinforcement learning exacerbates gradient interference, often leading to conservative averaging and selective failures on *high-dynamic behaviors* [12, 11, 9, 30]. Together, these factors cause tracking fidelity to collapse as motion diversity and difficulty increase.

The second barrier is the **physical executability bottleneck** that emerges at deployment. Even when high-fidelity tracking is achieved in simulation, transferring such behaviors to physical robots remains challenging. In prior humanoid learning pipelines [11, 58, 5, 23, 30, 55, 28, 14, 26, 49], actuation constraints during training are modeled primarily through joint position limits and simple effort bounds. Although these simplifications facilitate learning, they become insufficient in *high-dynamic motions*, where system behavior is dominated by unmodeled actuator nonlinearities [41], such as torque–speed characteristics and velocity-dependent torque losses, as well as power-related effects, including regenerative power phenomena, leading to rapid degradation of execution stability. As a result, fidelity that appears scalable in simulation may still fail to materialize on real robots.

Motivated by this analysis, we propose OMNIXTREME, a scalable training framework designed to explicitly address both barriers, with the goal of enabling a single policy to robustly control diverse and *high-dynamic* humanoid behaviors. To overcome the learning bottleneck, OMNIXTREME adopts a flow matching policy and performs specialist-to-unified generative pretraining via behavior cloning from a collection of motion specialists. This design decouples representation learning from optimization, scaling expressive capacity through a high-capacity generative policy while avoiding interference-heavy multi-motion reinforcement learning.

To overcome the physical executability bottleneck, OMNIXTREME introduces a residual reinforcement learning post-training refinement for execution under realistic actuation constraints, which become particularly critical in *high-dynamic motions*. Rather than relearning motion tracking, this stage refines the pretrained policy to respect real-world actuation constraints through actuation-aware modeling, refined domain randomization, and explicit penalties on power-related effects. This targeted refinement ensures that the scaled tracking policy remains physically executable under realistic hardware dynamics.

We validate OMNIXTREME through extensive simulation and real-robot evaluations on increasingly diverse and high-dynamic motion libraries. Beyond standard multi-motion

摘要—高保真运动追踪是检验泛化性、人类水平运动技能的终极试金石。然而，当前策略常遭遇"泛化壁垒"：随着运动库的多样性增加，追踪保真度不可避免地下降——尤其是在高动态运动的实际部署中。我们将此失败归因于两个叠加因素：多运动优化扩展中的学习瓶颈，以及实际驱动中产生的物理可执行性约束。为克服这些问题，我们引入***OMNIXTREME***，一个可扩展的框架，将通用运动技能学习与仿真到现实的物理技能精炼解耦。我们的方法采用具有高容量架构的流匹配策略，在不进行干扰性强的多运动强化学习优化的情况下扩展表示能力，随后进行驱动感知的精炼阶段，确保在物理硬件上实现稳健性能。大量实验表明，***OMNIXTREME***在各种高难度数据集上保持高保真追踪。在实际机器人上，统一策略成功执行多种极限运动，有效打破了高动态人形控制中保真度与可扩展性之间的长期权衡。

I. 引言

我们最终寻求的是具备可扩展、人类水平全身运动技能的通用型人形机器人。研究这种能力的一种自然且广泛使用的方法是高保真运动追踪，其中控制器必须精确地再现参考运动，同时在接触和干扰下保持动态稳定。高质量的追踪不仅仅是一个美学目标：它捕捉了支撑移动操作、表达性交互以及许多下游核心人形能力的全身协调和接触时机 [59, 50, 52, 1, 43, 47, 53]。

在过去的几年里，基于学习的运动追踪取得了显著进展：通过精心设计的目标函数和强化学习，策略可以以高精度追踪单个运动，包括舞蹈、空翻和武术等高动态行为 [57, 28, 14]。最近的工作 [5, 30, 9, 23, 11, 6, 12, 58, 55, 26] 在开发涵盖更广泛行为库的多运动控制器方面取得了重要进展。然而，一个反复出现的模式仍然存在：当我们扩展到涵盖不同风格、接触模式和时机模式的大型、更多样化的运动库时，运动追踪质量往往会下降。控制器变得保守和"平均"，在最难的运 动上失效，或在仿真到现实的转换中容易受到不可避免的微小偏差的影响而变得脆弱。这种退化在高动态运动中尤为明显，即使很小的追踪误差也可能迅速级联导致灾难性失败。这种长期存在的保真度-可扩展性权衡问题有效地限制了人形机器人运动控制中可实现的通用性水平，特别是在高动态范围内，这表明这是一种根本性限制而非孤立的工程问题 [11, 58, 14, 57]。

因此，一个核心问题随之产生：高保真运动追踪为何难以扩展，尤其是在真实人形机器人上？我们认为，这种困难源于当前模拟到真实训练流程中不同阶段出现的两个相互叠加的障碍。

第一个障碍是即使在模拟中也会出现的瓶颈学习。最近有数项研究 [58, 5, 23, 30, 55, 26, 12, 11]开始探索多动作人形跟踪，旨在提高超越单动作模仿的可扩展性。然而，现有方法仍然受限于表示和优化两方面。在表示方面，大多数方法依赖于相对简单的策略参数化，例如MLP演员 [58, 5, 23, 55, 26, 12, 11, 46]。当需要将观察映射到由不同行为和接触模式产生的异构动作目标时，这种参数化已被观察到随着数据多样性的增加而表现出有限的可扩展性 [33]。在优化方面，使用强化学习联合训练跨越多种动作的统一策略会加剧梯度干扰，通常导致对高动态行为的保守平均和选择性失败 [12, 11, 9, 30]。这些因素共同导致跟踪保真度随着动作多样性和难度增加而下降。

第二个障碍是部署时出现的物理可执行性瓶颈。即使模拟中实现了高保真追踪，将此类行为迁移到物理机器人上仍具挑战性。在先前的人形学习流程中，[11, 58, 5, 23, 30, 55, 28, 14, 26, 49]，训练时的驱动约束主要通过关节位置限制和简单的力矩界限进行建模。尽管这些简化有助于学习，但在高动态运动中却显得不足，此时系统行为主要由未建模的驱动器非线性特性 [41]，如力矩-速度特性以及速度相关的力矩损耗，以及与功率相关的效应（包括再生功率现象）主导，导致执行稳定性快速恶化。结果，在模拟中看似可扩展的保真度，在实际机器人上仍可能无法实现。

基于这项分析，我们提出了OMNIXTREME，这是一个可扩展的训练框架，旨在明确解决这两个障碍，目标是使单个策略能够稳健地控制多样化和高动态的人形机器人行为。为了克服学习瓶颈，OMNIXTREME采用流匹配策略，并通过从一组运动专家处进行行为克隆，进行专家到统一的生成预训练。这种设计将表示学习与优化解耦，通过高容量的生成策略扩展表达能力，同时避免干扰严重的多运动强化学习。

为了克服物理可执行性瓶颈，OMNIXTREME引入了针对真实驱动约束下的执行的后训练残差强化学习精炼，这在高动态运动中变得尤为重要。这个阶段不是重新学习运动跟踪，而是通过驱动感知建模、精细化的域随机化和对与功率相关的效应的显式惩罚，来精炼预训练策略，使其尊重现实世界的驱动约束。这种有针对性的精炼确保了扩展的跟踪策略在现实硬件动态下保持物理可执行性。

我们通过在日益多样化和高动态的运动库上进行广泛的仿真和真实机器人评估来验证 OMNIXTREME。超越标准多运动

benchmarks, we curate a set of extreme motions characterized by high speed, frequent contact transitions, and tight timing constraints, and evaluate OMNIXTREME across this full spectrum. As shown in Fig. 1, OMNIXTREME successfully executes a wide range of extreme behaviors on a Unitree G1 humanoid robot, including flips, acrobatics, and breakdancing where even minor deviations can rapidly cascade into failure. Together, these results constitute a stringent scalability stress test and challenge the prevailing assumption that tracking fidelity must collapse as motion diversity and difficulty increase.

Overall, our contributions are fourfold:

1) We present OMNIXTREME, a scalable training framework for high-fidelity humanoid motion tracking that explicitly tackles the fundamental scalability challenge in *high-dynamic* humanoid control.
2) We introduce a specialist-to-unified generative pretraining stage based on flow matching, enabling a unified policy to scale across heterogeneous and high-dynamic motions.
3) We propose an actuation-aware residual reinforcement learning post-training stage that refines the pretrained policy under realistic actuation constraints, ensuring physical executability.
4) We demonstrate through extensive simulation and real-world experiments that OMNIXTREME enables a single unified policy to robustly execute diverse and extreme motions, addressing the conventional fidelity–scalability trade-off, especially for high-dynamic motions.

II. RELATED WORK

A. Humanoid Whole-body Control and General Tracking

Recent research in humanoid whole-body control has demonstrated remarkable progress across diverse skills [62, 19, 15, 57, 36], including dance, fall recovery, and parkour. However, achieving both high-fidelity motion tracking and scalability across large and diverse motion libraries remains an open challenge. Frameworks such as ASAP [14] and BeyondMimic [28] demonstrate strong performance in high-quality imitation of individual motion clips, yet extending these approaches to increasingly large motion sets introduces additional optimization complexity. On the other hand, large-scale RL-based trackers including OmniH2O [11], ExBody2 [23], and GMT [5] show promising scalability, though maintaining precise motion fidelity under extensive skill coverage remains challenging. This tension is often reflected as a fidelity–scalability trade-off in practice. To address this issue, OMNIXTREME introduces a generative action representation and a specialist-to-unified optimization framework, enabling scalable learning while maintaining strong tracking precision across high-dynamic motion datasets.

B. Diffusion and Flow-based Action Modeling for Robotic Planning and Control

Diffusion and flow-based models [42, 16, 34, 38, 29, 33, 45, 44, 8] have shown strong capability in robot learning, leveraging iterative refinement and stochastic sampling to enhance robustness and diversity in robotic control and planning [33]. While early research focused on high-level trajectory planning or low-frequency visuomotor tasks [22, 18, 8, 48, 4], DiffuseLoco [20] takes a step to apply them to high-frequency quadruped control. To further enhance expressivity and robustness, recent works like Policy Decorator [54] and ResiP [2] introduce residual policy learning on arm-based robots, coupling frozen base models with refinement layers to handle covariate shift and precision bottlenecks in long-horizon assembly. However, given the vast skill space and inherent instability that distinguish humanoids from quadrupeds and manipulators, current effort such as BeyondMimic [28] focuses on guided control interfaces rather than the scalability and high-speed agility essential for high-dynamic humanoid motion tracking. Different from previous work, OMNIXTREME introduces a comprehensive training pipeline involving DAgger-based Flow Matching pretraining and residual post-training that pushes the boundaries of low-level scalability and agility, far surpassing the motion diversity and dynamic performance of previous approaches.

C. Actuation-aware Agile Robotic Control

Achieving agility remains a frontier in robotics [25, 27, 13, 24, 21, 32, 56, 17, 7, 61, 62, 49]. ACRL [41] leveraged actuator-constrained RL for high-speed quadrupedal locomotion, while Closing the Reality Gap [60] utilized a current-to-torque calibration and actuator dynamics modeling for dexterous five-finger manipulation. Despite these advancements in other morphologies, learning agile and actuation-aware control policies for humanoids remains an underexplored area. OMNIXTREME addresses this gap by integrating physics-informed motor modeling and actuation regularization, pushing the boundaries of agile humanoid performance under realistic hardware constraints.

III. METHODOLOGY

In this section, we present OMNIXTREME, a two-stage training framework for scalable, high-fidelity humanoid motor skill learning. The **Scalable Flow-based Pretraining** stage focuses on high-fidelity motion imitation and representation capacity acquisition. Specifically, we distill diverse expert behaviors from a collection of motion-specific expert policies into a single unified base policy using flow matching [29]. This generative pretraining stage establishes a shared tracking prior across heterogeneous motions, without relying on interference-prone joint multi-motion reinforcement learning.

To address the gap between simulation and real-world execution, we further introduce an **Actuation-Aware Post-Training** stage based on residual reinforcement learning. Rather than relearning motion tracking, a residual policy is trained to produce corrective actions that complement the pretrained flow matching base policy. This stage aligns the overall system with real-world actuation constraints while introducing substantially more aggressive domain randomization. Through this targeted refinement, the residual policy adapts the pretrained tracking behavior to realistic hardware

我们精心策划了一套极端运动，其特点是高速、频繁的接触转换和严格的时序约束，并在整个范围内评估OMNIXTREME。如图 1 所示，OMNIXTREME 成功地在 Unitree G1 机器人上执行了各种极端行为，包括翻转、杂技和霹雳舞，其中即使微小的偏差也可能迅速级联导致失败。这些结果共同构成了一个严格的可扩展性压力测试，挑战了运动多样性和难度增加时跟踪保真度必须崩溃的普遍假设。

Ov总体, ou贡献　u是针对　u折叠:

1) 我们提出了OMNIXTREME，一个用于高保真拟人机器人运动跟踪的可扩展训练框架，它明确地解决了高动态拟人机器人控制中的基本可扩展性挑战。2) 我们引入了一种基于流匹配的专业化到统一生成预训练阶段，使统一策略能够跨异构和高动态运动进行扩展。3) 我们提出了一种基于执行器感知的残差强化学习后训练阶段，该阶段在现实执行约束下优化预训练策略，确保物理可执行性。4) 通过广泛的模拟和真实世界实验，我们证明了OMNIXTREME使单个统一策略能够稳健地执行多样化和极端运动，解决了传统的保真度-可扩展性权衡问题，特别是对于高动态运动。

II. 相关工作

A. 人体全身控制与通用追踪

近年来，在人体全身控制领域的研究取得了显著进展，涵盖了舞蹈、跌倒恢复和跑酷等多种技能 [62, 19, 15, 57, 36],。然而，实现高保真运动追踪以及在大规模多样化运动库中的可扩展性仍然是一个开放性挑战。ASAP [14] 和BeyondMimic [28] 等框架在高质量单个运动片段的模仿方面表现出色，但将这些方法扩展到更大的运动集时会引入额外的优化复杂性。另一方面，OmniH2O [11], 、Ex-Body2 [23], 和GMT [5] 等大规模基于强化学习的追踪器显示出良好的可扩展性，但在广泛技能覆盖下保持精确的运动保真度仍然具有挑战性。这种矛盾在实践中通常表现为保真度与可扩展性之间的权衡。为解决这一问题，OMNIXTREME引入了一种生成式动作表示和专家到统一的优化框架，在实现可扩展学习的同时，能够保持在高动态运动数据集上的强追踪精度。

B. 基于扩散和流的动作建模用于机器人规划与控制

扩散模型和流模型 [42, 16, 34, 38, 29, 33, 45,44, 8] 在机器人学习领域已展现出强大能力，通过迭代优化和随机采样来增强机器人控制和规划的鲁棒性与多样性 [33]。虽然早期研究主要关注高级轨迹规划或低频视觉运动任务 [22, 18, 8, 48, 4],, DiffuseLoco [20] 则将它们应用于高频四足机器人控制。为进一步提升表达能力和鲁棒性，Policy Decorator [54] 和ResiP [2] 等近期工作在基于手臂的机器人上引入了残差策略学习，将冻结的基础模型与优化层结合，以处理长时程装配中的协变量偏移和精度瓶颈。然而，鉴于人形机器人与四足机器人和机械臂在技能空间广阔性和固有不稳定性上的差异，当前如BeyondMimic [28] 等研究主要关注引导式控制接口，而非高动态人形机器人运动跟踪所必需的可扩展性和高速敏捷性。与以往工作不同，OMNIXTREME引入了一套综合训练流程，包括基于DAgger的流匹配预训练和残差后训练，极大地推动了低层可扩展性和敏捷性的边界，远超以往方法在运动多样性和动态性能方面的表现。

C. 驱动感知的敏捷机器人控制

实现敏捷性仍然是机器人领域的挑战 [25, 27, 13, 24, 21, 32, 56, 17, 7, 61, 62, 49]。ACRL [41] 利用驱动器约束的强化学习实现高速四足运动，而Closing the Reality Gap [60]则采用电流到扭矩校准和驱动器动力学建模进行灵巧的五指操作。尽管其他形态领域取得了这些进展，但为拟人机器人学习敏捷和驱动感知的控制策略仍是一个尚未充分探索的领域。OMNIXTREME通过整合物理信息化的电机建模和驱动器正则化，弥补了这一空白，在现实硬件约束下将敏捷拟人机器人的性能边界推向了新的高度。

III. 方法论

在本节中，我们介绍了OMNIXTREME，一个用于可扩展、高保真拟人机器人电机技能学习的两阶段训练框架。可扩展的流式预训练阶段专注于高保真运动模仿和表示能力获取。具体而言，我们使用流匹配 [29] 将从一组特定运动的专家策略集合中提炼出的多样化专家行为，蒸馏到一个单一的统一基础策略中。这个生成式预训练阶段在异构运动之间建立了一个共享的跟踪先验，而无需依赖易受干扰的关节多运动强化学习。

为解决仿真与现实执行之间的差距，我们进一步引入一种基于残差强化学习的感知驱动后训练阶段。该阶段并非重新学习运动跟踪，而是训练一个残差策略，以生成补充预训练流匹配基础策略的纠正动作。这一阶段使整个系统与现实的驱动约束相协调，同时引入了大幅增强的领域随机化。通过这种针对性的改进，残差策略使预训练的跟踪行为适应了真实的硬件

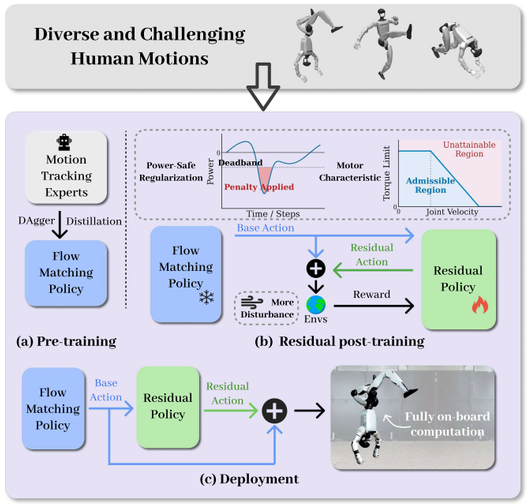


Fig. 2: **Overview of the OMNIXTREME.** (a) Pretraining phase: A unified base policy is trained via DAgger-based Flow Matching to aggregate diverse motion priors from different motion tracking experts. (b) Post-training phase: The base policy is frozen while a residual policy is optimized under stringent motor constraints, extensive domain randomization, and power-safety regularization to bridge the sim-to-real gap. (c) Onboard deployment: The whole inference pipeline is real-time and executed entirely onboard, facilitating robust and agile control in physical environments.

dynamics, improving physical executability and deployment robustness.

A. Scalable Flow-based Policy Pretraining

1) Problem Formulation: During the pretraining phase, we learn a flow-matching robot policy with Dataset Aggregation (DAgger)-based distillation [39, 45]. Specifically, we consider the observation space of $o = \{p, c, h\}$ covering: (i) robot proprioception p, including joint positions, velocities, base angular velocities, and previous actions; (ii) command c consisting of 6D torso orientation differences along with target joint positions and velocities from the reference motion; and (iii) history information h encompassing past proprioceptive states. Given a reference motion dataset $M = \{m_i\}_{i=1}^{M}$, our goal is to first learn expert policies $\Pi^i_{\text{expert}} = \{\pi^i_{\text{expert}}(a|o)\}_{i=1}^{M}$ for each reference motion, then distilling it into a flow-based general policy $\pi_\theta(a|o)$.

2) Expert Policy Learning: For expert policy training, we draw the reference motion dataset M from a combination of Unitree-retargeted LAFAN1 (LAFAN1) dataset [10], AMASS [31], MimicKit [35], and the Reallusion motion library [37], covering both diverse behavioral patterns and high-dynamic maneuvers. All reference motions are first retargeted to the Unitree G1 humanoid robot using GMR [55, 3]. Subsequently, we train each expert policy π^k_{expert} on the specific motion m_k via Proximal Policy Optimization (PPO) [40].

3) Flow-matching Policy Learning: We learn the flow-matching robot policy with DAgger by first rolling out the current flow-based policy $\pi_\theta(a|o)$ in the simulator and collecting a trajectory of visited states $\{o_1, \cdots, o_N\}$ given reference motion dataset M. For each visited state o, we obtain the expert action a_{expert} by querying the corresponding expert policy. The flow-based model then learn to recover the expert action a_{expert} from noised action by optimizing:

$$\mathcal{L}_{\text{FM}}(\theta) = \mathbb{E}_{t,\epsilon,a_{\text{expert}}}\left[\|v_\theta(a_t, t, o) - (\epsilon - a_{\text{expert}})\|^2\right], \quad (1)$$

where $a_t = (1-t)a_{\text{expert}} + t\epsilon$ is the noised action interpolated between expert action a_{expert} and random noise $\epsilon \sim \mathcal{N}(0, I)$ depending on flow timestep $t \in [0, 1]$. This objective learns a velocity field $v_\theta(a_t, t, o)$ to predict the target velocity $u = \epsilon - a_{\text{expert}}$, learning the denoising directions at each flow timestep [29]. During the optimization process, the timestep t is sampled from a Beta distribution, $t \sim \text{Beta}(\alpha, \beta)$, to focus the learning process on specific regions of the probability path, thereby enhancing convergence and trajectory refinement. With the velocity field v_θ, we can generate action a_0 from random noise $a_1 \sim \mathcal{N}(0, I)$ by integrating v_θ from $t = 1$ to $t = 0$ via the foward Euler rule:

$$a_{t-\frac{1}{D}} = a_t - \frac{1}{D} v_\theta(a_t, t, o), \quad (2)$$

where D is the number of integration or denoising steps controlling the approximation accuracy. By iteratively rolling out trajectories and supervising them with expert actions using Eq. (1), we learn π_θ as a general policy to map the current observation to appropriate actions. The full training procedure is illustrated in Fig. 2(a) and detailed in Alg. 1.

4) Fidelity-Preserving Randomization and Noise: To maintain a high degree of motion expressivity while ensuring physical stability, we implement a conservative randomization and noise strategy, as detailed in Tab. I, during both the teacher training and pretraining phases. By utilizing moderate noise levels and domain randomization, we prevent the performance collapse often induced by excessive stochasticity. This ensures that the agent accurately captures the underlying physical dynamics, resulting in a flow matching policy that possesses foundational sim-to-real robustness and the predictive certainty necessary for real-world deployment.

B. Actuation-Aware Post-training Phase

1) Residual Policy Modeling: While the pretrained flow matching base policy provides a robust and unified behavioral foundation, it encounters performance gaps when facing real-world physics. To better account for this gap and enable smooth sim-to-real transfer, we propose a post-training refinement stage using a lightweight MLP-based residual-corrective learning. Specifically, we learn the residual correction policy π_ϕ on top of the frozen pretrained policy π_θ by generating the refined action $a = a_{\text{flow}} + a_{\text{res}}$ and supervising it with cumulative rewards via PPO, detailed in the Appendix.

In particular, the observation space for the residual actor and critic incorporates robot proprioception, motion command,

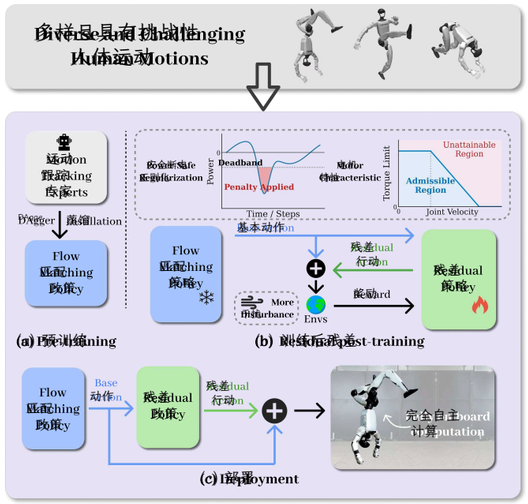


图2：OMNIXTREME 概览。(a) 预训练阶段：通过基于 DAgger 的流匹配方法训练一个统一的基策略，以从不同的运动跟踪专家那里聚合多样的运动先验。(b) 训练后阶段：基策略被冻结，同时一个残差策略在严格的运动约束、广泛的领域随机化和功率安全正则化下进行优化，以弥合仿真到现实的差距。(c) 船载部署：整个推理流程是实时的，并且完全在船载上执行，便于在物理环境中实现稳健和敏捷的控制。

动态特性，提高物理可执行性和部署鲁棒性。

A. 可扩展的基于流的策略预训练

1) 问题表述：在预训练阶段，我们学习一个基于数据集聚合（DAgger）蒸馏的流匹配机器人策略 [39, 45]。具体来说，我们考虑的观测空间 $o = \{p, c, h\}$ 涵盖：(i) 机器人本体感觉 p，包括关节位置、速度、基座角速度和先前动作；(ii) 命令 c，由6D躯干方向差以及参考运动中的目标关节位置和速度组成；(iii) 历史信息 h，包含过去的本体感觉状态。给定一个参考运动数据集 $M = \{m_i\}_{i=1}^{M}$，我们的目标首先是学习每个参考运动的专家策略 $\Pi^i_{\text{expert}} = \{\pi^i_{\text{expert}}(a|o)\}_{i=1}^{M}$，然后将它蒸馏成一个基于流的通用策略 $\pi_\theta(a|o)$。

2) 专家策略学习：对于专家策略训练，我们从 Unitree-重定向的 LAFAN1 (LAFAN1) 数据集 M、AMASS [31]，MimicKit [35]，以及覆盖多样化行为模式和高速动态动作的 Reallusion 动作库 [37]，的组合中提取参考运动数据集 [10],。所有参考运动首先使用 GMR [55, 3]重定向到 Unitree G1 机器人上。随后，我们通过近端策略优化 (PPO) [40]在特定运动 m_k 上训练每个专家策略π^k_{expert}。

3) 流匹配策略学习：我们通过 DAgger 学习流匹配机器人策略，首先在模拟器中部署当前基于流的策略 $\pi_\theta(a|o)$，并根据参考运动数据集 M收集访问状态的轨迹 $\{o_1, \cdots, o_N\}$。对于每个访问状态 o，我们通过查询相应的专家策略获得专家动作a_{expert} 。基于流的模型随后通过优化来学习从噪声动作中恢复专家动作a_{expert} ：

$$\mathcal{L}_{\text{FM}}(\theta) = \mathbb{E}_{t,\epsilon,a_{\text{expert}}}\left[\|v_\theta(a_t, t, o) - (\epsilon - a_{\text{expert}})\|^2\right], \quad (1)$$

其中 $a_t = (1-t)a_{\text{expert}} + t\epsilon$ 是在专家动作a_{expert}和随机噪声 $\epsilon \sim \mathcal{N}(0, I)$之间根据流时间步长 $t \in [0, 1]$插值的噪声动作。这个目标学习一个速度场 $v_\theta(a_t, t, o)$来预测目标速度 $u = \epsilon - a_{\text{expert}}$，并在每个流时间步长 [29]学习去噪方向。在优化过程中，时间步长t 从Beta分布 $t \sim \text{Beta}(\alpha, \beta)$中采样，以将学习过程集中在概率路径的特定区域，从而提高收敛性和轨迹优化。通过速度场 v_θ，我们可以通过从 $t = 1$ 到 $t = 0$ 通过正向欧拉规则积分 v_θ，从随机噪声 $a_1 \sim \mathcal{N}(0, I)$生成动作$a_0$：

$$a_{t-\frac{1}{D}} = a_t - \frac{1}{D} v_\theta(a_t, t, o), \quad (2)$$

D 是控制近似精度的积分或去噪步数。通过迭代展开轨迹并用专家动作（使用公式(1)）对其进行监督，我们学习 π_θ 作为将当前观察映射到适当动作的通用策略。完整的训练过程如图2(a)所示，并在算法1中详细说明。

4) 保持高运动表现力的保真度随机化和噪声：为了在确保物理稳定性的同时保持高度的运动表现力，我们在教师训练和预训练阶段都实施了保守的随机化和噪声策略，具体见表I。通过使用适度的噪声水平和领域随机化，我们防止了过度随机性通常引起的性能崩溃。这确保了智能体准确捕捉了潜在的物理动态，从而产生具有基础仿真到现实鲁棒性和必要预测确定性的流匹配策略，使其能够部署于现实世界。

B. 驱动感知的离线训练阶段

1) 残差策略建模：虽然预训练的流匹配基础策略提供了一个鲁棒且统一的行为基础，但在面对现实物理时仍会遇到性能差距。为了更好地弥补这一差距并实现平滑的仿真到现实迁移，我们提出使用轻量级基于MLP的残差校正学习进行离线训练优化。具体而言，我们在冻结的预训练策略 π_θ 之上学习残差校正策略π_ϕ ，通过生成优化动作 $a = a_{\text{flow}} + a_{\text{res}}$ 并通过PPO累积奖励进行监督，具体见附录。

特别是，残差演员和评价者的观测空间包含了机器人本体感觉、运动指令，

TABLE I: Configurations for noise, domain randomization, and termination thresholds during pre-training and post-training phases. Here $\pm x$ denotes $[-x, x]$.

Parameter Item	Moderate	Aggressive
Noise and Domain Randomization		
Joint Position (rad)	± 0.01	± 0.01
Joint Velocity (rad/s)	± 0.5	± 0.5
Angular Velocity (rad/s)	± 0.2	± 0.2
Torso 6D Rotation (rad)	± 0.05	± 0.05
Base CoM Offset (m)	x: ± 0.025, y, z: ± 0.05	x: ± 0.025, y, z: ± 0.05
Static Friction	$[0.3, 1.6]$	$[0.3, 1.6]$
Dynamic Friction	$[0.3, 1.2]$	$[0.3, 1.2]$
Action Delay (ms)	$[0, 15]$	$[\mathbf{5}, \mathbf{10}]$
Coefficient of Restitution	None	$[\mathbf{0.0}, \mathbf{0.5}]$
Default Calib. (rad)	± 0.01	± 0.01
Init. Pose (rad)	± 0.1	$\pm \mathbf{0.15}$
Init. Lin. Vel. (m/s)	xy: ± 0.5, z: ± 0.2	$\mathbf{xy}$: $\pm \mathbf{0.75}$, $\mathbf{z}$: $\pm \mathbf{0.3}$
Init. Ang. Vel. (rad/s)	RP: ± 0.52, Y: ± 0.78	$\mathbf{RP}$: $\pm \mathbf{0.78}$, $\mathbf{Y}$: $\pm \mathbf{1.17}$
Push Frequency (s)	$1.0 - 3.0$	$1.0 - 3.0$
Push Lin. Vel. (m/s)	xy: ± 0.5, z: ± 0.2	xy: ± 0.5, z: ± 0.2
Push Ang. Vel. (rad/s)	RP: ± 0.52, Y: ± 0.78	RP: ± 0.52, Y: ± 0.78
Terrain Surface / Step (m)	None	$[\mathbf{0}, \mathbf{0.01}]/\mathbf{0.01}$
Termination Thresholds		
Torso Pos. Z / Ori. Error	0.25m / 0.8rad	$\mathbf{0.375m}/\mathbf{1.2rad}$
End-Effector Z-Error (m)	0.25	$\mathbf{0.375}$

and the current base action a_{flow}. Within the proprioceptive state, the residual policy observes the previous refined action, whereas the flow matching base policy remains conditioned on the previous flow-based action.

2) Actuation-aware Physical Constraint Modeling: To explicitly account for real-world actuation effects, we train the residual policy using environments that incorporate realistic actuation-aware physical constraints and domain randomization, as shown in Fig. 2(b). The actuation-aware physical modeling is detailed as follows:

a) Aggressive Domain Randomization: We substantially increase the range for domain randomization by up to 50% on common domain randomization settings, including initial pose noise, force disturbances magnitude, angular velocity, *etc*., as detailed in Tab. I. We randomize the terrain by adding surface noise and placing vertical steps randomly in the scene. Crucially, we relax the termination thresholds by 1.5× from their base values (*e.g*., orientation error from 0.8 to 1.2 rad). This relaxation allows the residual policy to explore and correct for large-deviation but recoverable states that would otherwise be prematurely terminated.

b) Power-Safe Actuation Regularization: In practice, highly dynamic motions can induce large transient braking loads that are not explicitly regulated in standard training pipelines. To address the issue, we introduce an explicit penalty on excessive negative joint mechanical power to mitigate aggressive motor braking that can trigger overcurrent protection or thermal stress on real robots. Specifically, we use the instantaneous mechanical power $P = \tau \cdot \omega$ calculated from the applied joint torque τ and angular velocity ω as a critical policy for actuator safety. We penalize the negative power beyond a predefined deadband to suppress large regenerative

Algorithm 1 Flow-based Pretraining and Inference

Training: Distill Flow Matching Policy with DAgger
Input: Teacher policy set $\pi_{teacher}$, Flow matching policy π_θ, Motion dataset $\mathcal{M}$, Replay buffer $\mathcal{D}$
repeat
  $\mathcal{D} \leftarrow \emptyset$ ▷ On-policy reset: Clear buffer for the new iteration
  Sample motion $m \sim \mathcal{M}$ and select teacher $\pi^m_{teacher}$
  Rollout π_θ in simulator conditioned on m to obtain states $s_{1:T}$
  for $t = 1$ to T **do**
    $a_{expert,t} \leftarrow \pi^m_{teacher}(s_t)$ ▷ Expert labeling
    $\mathcal{D} \leftarrow \mathcal{D} \cup \{(s_t, m, a_{expert,t})\}$ ▷ Aggregate data
  end for
  Flow Matching Optimization:
  for each gradient step **do**
    Sample $(s, a_{expert}) \sim \mathcal{D}$
    Sample $t \sim \text{Beta}(\alpha, \beta)$ and $\epsilon \sim \mathcal{N}(0, I)$
    Construct probability path: $x_t = (1-t)a_{expert} + t\epsilon$
    Compute target velocity: $u_t = \epsilon - a_{expert}$
    Update θ using gradient descent on $\|v_\theta(x_t, t, c) - u_t\|^2$
  end for
until convergence

Inference: Action Sampling with Euler Integration
Input: Trained velocity field v_θ, Concatenated condition c, Number of steps N
Set step size $\Delta t = 1/N$
Initialize $x_1 \sim \mathcal{N}(0, I)$ ▷ Start from Gaussian noise
for $k = 0$ to $N - 1$ **do**
  $t = 1 - k \cdot \Delta t$ ▷ Reverse time from 1 to 0
  $v_t \leftarrow v_\theta(x_t, t, c)$
  $x_{t-\Delta t} \leftarrow x_t - v_t \cdot \Delta t$ ▷ Reverse-time Euler step to obtain x_0
end for
return $a = x_0$ ▷ Execute final reconstructed action

braking events for each joint:

$$\mathcal{L}_{\text{neg-power}} = \sum_{j \in \mathcal{J}} \left(\frac{\max(-P_j - P_{\text{db}}, 0)}{K} \right)^2, \qquad (3)$$

where P_j, P_{db} denotes power for joint j and the deadband threshold, respectively. K is a normalizing constant. In practice, this term is selectively applied to the knee joints in the context of high-dynamic motions(*e.g*., backflips), as these joints are particularly prone to high braking loads during impacts and recovery phases.

c) Actuation-Aware Torque–Speed Constraints: A major source of sim-to-real discrepancy stems from the oversimplification of actuator modeling, whereas standard torque clipping techniques neglect velocity-dependent constraints imposed by back-electromotive force and physical power limits. This omission leads to a significant sim-to-real gap when performing

表 I：预训练和后训练阶段噪声、域随机化和终止阈值配置。此处 $\pm x$ 表示 $[-x, x]$。

参数项	中等	激进
噪声和域随机化		
关节位置 (rad)	± 0.01	± 0.01
关节速度 (rad/s)	± 0.5	± 0.5
角速度 (rad/s)	± 0.2	± 0.2
躯干6D旋转 (rad)	± 0.05	± 0.05
基座CoM偏移 (m)	x: ± 0.025, y, z: ± 0.05	x: ± 0.025, y, z: ± 0.05
静摩擦力	$[0.3, 1.6]$	$[0.3, 1.6]$
动态摩擦	$[0.3, 1.2]$	$[0.3, 1.2]$
动作延迟 (ms)	$[0, 15]$	$[\mathbf{5}, \mathbf{10}]$
恢复系数	None	$[\mathbf{0.0}, \mathbf{0.5}]$
默认校准 (rad)	± 0.01	± 0.01
初始姿态 (rad)	± 0.1	$\pm \mathbf{0.15}$
初始线速度 (m/s)	xy: ± 0.5, z: ± 0.2	$\boldsymbol{xy:}$ $\pm \mathbf{0.75}$, $\boldsymbol{z:}$ $\pm \mathbf{0.3}$
初始角度速度 (rad/s)	RP: ± 0.52, Y: ± 0.78	$\boldsymbol{RP:}$ $\pm \mathbf{0.78}$, $\boldsymbol{Y:}$ $\pm \mathbf{1.17}$
推力频率 (s)	$1.0 - 3.0$	$1.0 - 3.0$
推力线性速度 (m/s)	xy: ± 0.5, z: ± 0.2	xy: ± 0.5, z: ± 0.2
推力角速度 (rad/s)	RP: ± 0.52, Y: ± 0.78	RP: ± 0.52, Y: ± 0.78
地形表面 / 步伐 (m)	None	$[\mathbf{0}, \mathbf{0.01}]/\mathbf{0.01}$
终止阈值		
躯干位置 Z / 方向误差	0.25m / 0.8rad	$\boldsymbol{0.375m}/\mathbf{1.2rad}$
末端执行器 Z-误差 (m)	0.25	$\mathbf{0.375}$

以及当前的基础动作 a_{flow}。在本体感觉状态下，残差策略观测前一步优化的动作，而流匹配基础策略则仍依赖于前一步基于流的动作。

2) 驱动感知物理约束建模：为了明确考虑现实世界的驱动效应，我们使用包含真实驱动感知物理约束和领域随机化的环境来训练残差策略，如图2(b)所示。驱动感知物理建模的细节如下：

a) 激进领域随机化：我们在常见的领域随机化设置上，将领域随机化的范围大幅增加高达50%，包括初始姿态噪声、力扰动幅度、角速度等，如表I详细说明。我们通过添加表面噪声和在场景中随机放置垂直台阶来随机化地形。关键在于，我们将终止阈值从基准值放宽了1.5×（例如，方向误差从0.8rad放宽到1.2rad）。这种放宽允许残差策略探索和纠正那些原本会被过早终止的大偏差但可恢复的状态。

b) 功率安全驱动正则化：在实际应用中，高动态运动可能引发较大的瞬时制动力，而标准训练流程中并未明确对此进行调节。为解决这一问题，我们引入对过度负向关节机械功率的显式惩罚，以减轻可能触发机器人过流保护或热应力的激进电机制动。具体而言，我们使用从施加的关节扭矩 τ 和角速度 ω 计算出的瞬时机械功率 $P = \tau \cdot \omega$ 作为驱动器安全的关键策略。我们惩罚超出预设死区的负向功率，以抑制大额再生

算法1 基于流的预训练与推理 1：训练：使用DAgger蒸馏流匹配策略 2：输入：教师策略集 $\pi_{teacher}$、流匹配策略π_θ、运动数据集 $\mathcal{M}$、重放缓冲区 $\mathcal{D}$ 3：重复 4：$\mathcal{D} \leftarrow \emptyset$ ▷ 按策略重置：清除新迭代缓冲区 5：采样运动 $m \sim \mathcal{M}$ 并选择教师 $\pi^m_{teacher}$ 6：在模拟器中基于 m 运行 π_θ 以获取状态 $s_{1:T}$ 7：对于 $t = 1$ 到 T 执行 8：$a_{expert,t} \leftarrow \pi^m_{teacher}(s_t)$ ▷ 专家标注 9：$\mathcal{D} \leftarrow \mathcal{D} \cup \{(s_t, m, a_{expert,t})\}$ ▷聚合数据 10：结束循环 11：流匹配优化： 12：对于每个梯度步执行 13：采样 $(s, a_{expert}) \sim \mathcal{D}$ 14：采样 $t \sim \text{Beta}(\alpha, \beta)$ 和 $\epsilon \sim \mathcal{N}(0, I)$ 15：构建概率路径：$x_t = (1-t)a_{expert} + t\epsilon$ 16：计算目标速度：$u_t = \epsilon - a_{expert}$ 17：使用梯度下降在 $\|v_\theta(x_t, t, c) - u_t\|^2$ 上更新 θ 18：结束循环 19：直到收敛 20：推理：使用欧拉积分进行动作采样 21：输入：训练好的速度场 v_θ、拼接条件c、步数 N 22：设置步长 $\Delta t = 1/N$ 23：初始化 $x_1 \sim \mathcal{N}(0, I)$ ▷从高斯噪声开始 24：对于 $k = 0$ 到 $N - 1$ 执行 25：$t = 1 - k \cdot \Delta t$ ▷从1反向至0 26：$v_t \leftarrow v_\theta(x_t, t, c)$ 27：$x_{t-\Delta t} \leftarrow x_t - v_t \cdot \Delta t$ ▷反向时间欧拉步以获取 x_0 28：结束循环 29：返回 $a = x_0$ ▷执行最终重建动作

每个关节的制动事件：

$$\mathcal{L}_{\text{neg-power}} = \sum_{j \in \mathcal{J}} \left(\frac{\max(-P_j - P_{\text{db}}, 0)}{K} \right)^2, \qquad (3)$$

其中 P_j, P_{db} 分别表示关节 j 的功率和死区阈值。K 是一个归一化常数。在实际应用中，该项会根据高动态运动（例如后空翻）的上下文选择性地应用于膝关节，因为这些关节在撞击和恢复阶段特别容易承受高制动负载。

c) 驱动器感知的扭矩-速度约束：模拟到现实的差异主要源于执行器建模的过度简化，而标准的扭矩裁剪技术忽略了反电动势和物理功率限制所施加的与速度相关的约束。这种遗漏在进行高动态运动时会导致显著的模拟到现实差距。

high-dynamic motions. To bridge this gap, we integrate a realistic torque-speed operating envelope directly into the simulation, dynamically deriving torque limits based on the instantaneous alignment of torque and angular velocity:

$$\tau_{\max,0} = \begin{cases} \tau_{y1}, & v \cdot \tau_{\text{in}} > 0, \\ \tau_{y2}, & v \cdot \tau_{\text{in}} \leq 0. \end{cases} \tag{4}$$

The admissible torque is then defined as a monotonically decreasing function of joint velocity magnitude:

$$\tau_{\text{clipped}}(v) = \begin{cases} \tau_{\max,0}, & |v| < v_{x1}, \\ \tau_{\max,0}\left(1 - \dfrac{|v| - v_{x1}}{v_{x2} - v_{x1}}\right), & v_{x1} \leq |v| \leq v_{x2}, \\ 0, & |v| > v_{x2}. \end{cases} \tag{5}$$

The commanded torque is finally clipped to this admissible range before being applied to the joint, which ensures that the simulator never samples torque commands that are physically unattainable for the real actuators.

In addition to torque-speed limits, we further model actuator-level internal losses through a nonlinear friction term applied after torque clipping,

$$\tau_{\text{applied}} = \tau_{\text{clipped}} - \left(\mu_s \tanh\left(\frac{v}{v_{\text{act}}}\right) + \mu_d v\right). \tag{6}$$

The smoothed Coulomb component captures the transition from static to kinetic friction, while the viscous term accounts for velocity-dependent dissipation and provides additional damping. The parameters μ_s, v_{act}, and μ_d are constants.

Overall, this structured refinement stage yields controllers that are simultaneously safer, more robust to large disturbances, and more faithfully aligned with real-world actuator dynamics, thereby enabling reliable deployment on robots.

C. Real World Deployment

The integrated real-world deployment pipeline is illustrated in Fig. 2(c). In the deployment phase, we leverage the pelvis IMU as the primary orientation source and compute the torso rotation through Forward Kinematics (FK). To ensure minimal control latency, the entire computational pipeline—including FK-based state estimation, the base flow matching policy, and the residual policy—is optimized and executed via TensorRT. This integrated pipeline achieves an end-to-end inference latency of about 10ms on the Unitree G1's onboard Orin NX. Such optimization enables the robot to execute high-quality motion tracking at a consistent 50Hz frequency in complex physical environments.

IV. Experiments

We present extensive experiments in simulation and on physical robots to evaluate the scalability of our proposed OmniXtreme system as motion libraries grow in diversity and difficulty. Our experiments are organized around the following key questions:

Q1: Scalable high-fidelity tracking. Compared to prior multi-motion baselines, can our approach maintain high-fidelity tracking at scale, both in simulation and on real robots, without collapsing under representation and optimization challenges?

Q2: Fidelity–scalability trade-off (OmniXtreme v.s. from-scratch RL controllers). As motion diversity and difficulty increase, how does tracking performance degrade for from-scratch multi-motion reinforcement learning controllers, and to what extent can our approach extend the scalability frontier?

Q3: Capacity scaling with flow-based (OmniXtreme v.s. MLP-based controllers). Does increasing model capacity improve large-scale multi-motion tracking performance, and does generative pretraining via flow matching enable stronger and more stable scaling behavior than conventional MLP-based motion tracking controllers?

Q4: Real-world executability and robustness. How do aggressive domain randomization, actuation-aware modeling, and power-aware safety mechanisms individually and jointly affect sim-to-real transfer and real-world execution success?

Q5: Qualitative whole-body capability. Beyond scalar tracking metrics, can OmniXtreme demonstrate agile and versatile whole-body behaviors across diverse motion styles and dynamic contact patterns?

Together, these questions probe the scalability and robustness of OmniXtreme by disentangling the roles of generative pretraining for representation and capacity scaling, and residual post-training for real-world executability.

A. Experimental Setup

1) Motion Libraries: We construct our motion libraries following a two-tier design. First, we use the full LAFAN1 [10], which has been widely adopted in prior multi-motion tracking work and serves as a standard benchmark for evaluating scalability under stylistic and temporal diversity.

Second, to evaluate and push the limit of extreme humanoid motions, we curate about 60 highly challenging motions selected from LAFAN1 [10], AMASS [31], MimicKit [35], and Reallusion [37]. These motions exhibit substantially higher dynamic intensity, frequent contact transitions, and tight timing constraints, as shown in Fig. 1(a). We collectively refer to this curated set as the XtremeMotion dataset.

Together, LAFAN1 and XtremeMotion form a motion library that spans both standard multi-motion benchmarks and extreme behaviors that probe the limits of fidelity, robustness, and real-world executability.

2) Baselines: We compare against two families of strong baselines designed for multi-motion tracking.

(a) Specialist-to-Unified MLP Distillation. This class of methods [58] first trains per-motion (or per-cluster) specialist policies and then distills them into a single unified MLP tracking policy. Relying on supervised distillation, they benefit from relatively stable and straightforward optimization, but are limited by the representational capacity of the MLP policy.

(b) From-scratch Multi-motion Reinforcement Learning. This class [11, 5, 23, 58, 30, 28] directly trains a single unified tracking policy from scratch using reinforcement learning across all motions, but often suffers from gradient interference

为弥补这一差距，我们将真实的扭矩-速度工作包络直接集成到模拟中，根据扭矩和角速度的瞬时对齐动态推导扭矩限制：

$$\tau_{\max,0} = \begin{cases} \tau_{y1}, & v \cdot \tau_{\text{in}} > 0, \\ \tau_{y2}, & v \cdot \tau_{\text{in}} \leq 0. \end{cases} \tag{4}$$

容许扭矩被定义为关节速度幅值的单调递减函数：

$$\tau_{\text{clipped}}(v) = \begin{cases} \tau_{\max,0}, & |v| < v_{x1}, \\ \tau_{\max,0}\left(1 - \dfrac{|v| - v_{x1}}{v_{x2} - v_{x1}}\right), & v_{x1} \leq |v| \leq v_{x2}, \\ 0, & |v| > v_{x2}. \end{cases} \tag{5}$$

指令扭矩最终被裁剪到这个容许范围内后才施加到关节上，这确保了模拟器永远不会采样对真实执行器物理上无法达到的扭矩指令。

除了扭矩-速度限制外，我们还通过在扭矩裁剪后应用非线性摩擦项来进一步对执行器级别的内部损耗进行建模，

$$\tau_{\text{applied}} = \tau_{\text{clipped}} - \left(\mu_s \tanh\left(\frac{v}{v_{\text{act}}}\right) + \mu_d v\right). \tag{6}$$

平滑的库伦分量捕获了从静摩擦到动摩擦的过渡，而粘性项则考虑了速度相关的耗散并提供了额外的阻尼。参数 μ_s、v_{act}和 μ_d 是常数。

总体而言，这个结构化精炼阶段产生了同时更安全、对大扰动更鲁棒、并且更忠实地与真实世界执行器动力学相一致的控制器，从而能够实现在机器人上的可靠部署。

C. 真实世界部署

集成化的实际部署流程如图2(c)所示。在部署阶段，我们利用骨盆IMU作为主要的方向源，并通过正向运动学(FK)计算躯干旋转。为确保最小控制延迟，整个计算流程——包括基于FK的状态估计、基础流匹配策略和残差策略——都通过TensorRT进行优化和执行。该集成化流程在Unitree G1的板载Orin NX上实现了约10ms的端到端推理延迟。这种优化使机器人在复杂的物理环境中能够以50Hz的稳定频率执行高质量的轨迹跟踪。

IV. 实验

我们在仿真和物理机器人上进行了广泛的实验，以评估OMNIXTREME系统在运动库随多样性和难度增长时的可扩展性。我们的实验围绕以下关键问题展开：

Q1：可扩展的高保真追踪。与先前的多运动基线相比，我们的方法能否在模拟和真实机器人上保持高保真追踪，同时不因表示和优化挑战而崩溃？

Q2：保真度-可扩展性权衡（OMNIXTREME与从头开始RL控制器）。随着运动多样性和难度的增加，从头开始的运动强化学习控制器的跟踪性能如何退化，以及我们的方法能在多大程度上扩展可扩展性边界？

Q3：基于流的（OMNIXTREME 与基于 MLP 的控制器）的容量扩展。增加模型容量是否提高了大规模多动作跟踪性能，以及通过流匹配的生成式预训练是否比传统的基于 MLP 的运动跟踪控制器带来更强和更稳定的扩展行为？

Q4：现实世界的可执行性和鲁棒性。激进的场景随机化、驱动感知建模和功率感知安全机制在单独和共同作用下如何影响仿真到现实的迁移和现实世界执行的成功？

Q5：定性全身能力。除了标量跟踪指标，OMNIXTREME 是否能在不同运动风格和动态接触模式中展示敏捷和通用的全身行为？

这些问题的共同点是，通过将生成式预训练在表示和容量扩展中的作用与残余后训练在现实世界可执行性中的作用分离，来探究 OMNIXTREME 的可扩展性和鲁棒性。

A. 实验设置

1）运动库：我们按照双层设计构建运动库。首先，我们使用广泛用于先前多动作跟踪工作的完整 LAFAN 1 [10],，它作为评估在风格和时间多样性下可扩展性的标准基准。

其次，为了评估和突破极端人形动作的极限，我们从 LAFAN1 [10], AMASS [31], MimicKit [35], 和 Reallusion [37] 中精选了约60个极具挑战性的动作。这些动作表现出显著更高的动态强度、频繁的接触转换和严格的时序约束，如图1(a)所示。我们将这个精选的数据集统称为XtremeMotion数据集。

一起，LAFAN1 和 XtremeMotion 形成一个涵盖标准多动作基准和测试保真度、鲁棒性和现实世界可执行性的极限行为的动作库。

2) 基线：我们与两种为多运动跟踪设计的强基线进行了比较。

(a) 专家网络到统一MLP蒸馏。这类方法 [58] 首先训练每个动作（或每个簇）的专家策略，然后将它们蒸馏成一个统一的MLP跟踪策略。依靠监督蒸馏，它们受益于相对稳定和直接的优化，但受限于MLP策略的表示能力。

(b) 从零开始的多动作强化学习。这类方法 [11, 5, 23, 58, 30, 28]直接使用强化学习在所有动作上从头训练一个单一的统一跟踪策略，但经常受到梯度干扰

and conservative averaging as motion diversity and difficulty increase.

B. Evaluation Metrics

The policy is evaluated through simulated rollouts of motion tracking to extract performance metrics. The primary metric is the **success rate (Succ)**, where an episode is deemed unsuccessful if the humanoid deviates beyond a predefined threshold from the reference motion or becomes unstable. We additionally report the **root-relative mean per-joint position error (MPJPE)** (mm), as well as discrepancies in joint-space **velocity** (Δvel) and **acceleration** (Δacc), to quantify kinematic accuracy and physical fidelity.

On physical robots, we evaluate performance using deployment-oriented metrics, including **skill-level success rates** and qualitative assessments of motion fidelity for high-dynamic behaviors.

C. Scalable high-fidelity tracking (Q1)

In this section, we study whether high-fidelity humanoid motion tracking can be preserved by OMNIXTREME as motion libraries scale in diversity and difficulty. We compare OMNIXTREME with specialist-to-unified MLP distillation and from-scratch multi-motion reinforcement learning under matched model capacity and identical training data. All methods are trained on the same combined motion library (LAFAN1 + XtremeMotion) and evaluated on three test sets: the full motion library, the high-dynamic XtremeMotion subset, and an unseen motion set (randomly sampled from retargeted AMASS).

Simulation. As summarized in Tab. II, **OMNIXTREME consistently outperforms both baselines across all simulation metrics.** The gap becomes substantially larger on XtremeMotion and unseen motions, where baseline methods exhibit reduced success rates and increased tracking errors as motion difficulty increases. This indicates that OMNIXTREME preserves tracking fidelity as motion diversity and difficulty scale, rather than degrading under increased complexity.

Real world. We further deploy OMNIXTREME on a Unitree G1 humanoid robot using motions drawn from XtremeMotion. For clarity of presentation, motions are grouped into representative skill categories based on shared dynamic structure and contact patterns. A trial is considered successful if the motion is executed without manual intervention or safety-triggered termination. As shown in Tab. III, across 157 real-world trials spanning 24 distinct high-dynamic motions, **OMNIXTREME achieves consistently high success rates across diverse skill categories**, including flips, acrobatics, breakdancing, and martial-arts-style motions. These results demonstrate that the scalability gains observed in simulation translate to robust and physically executable behaviors on real hardware.

D. Fidelity–scalability trade-off (Q2)

To characterize the fidelity–scalability trade-off in multi-motion tracking, we progressively scale motion diversity by training on an expanding set of motions drawn from XtremeMotion, and analyze how different training paradigms respond under the same evaluation protocol.

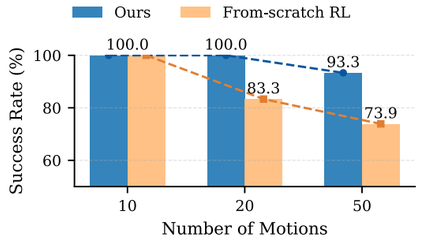


Fig. 3: **Fidelity–scalability trade-off.** Tracking success rate as we progressively scale motion diversity and difficulty, while evaluating all policies on a fixed set of the first 10 motions.

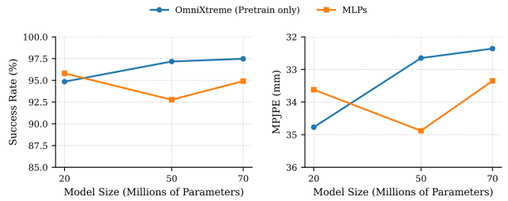


Fig. 4: **Capacity scaling.** Tracking fidelity and robustness as a function of model capacity. OMNIXTREME benefit more strongly from scaling, while conventional MLP controllers saturate earlier.

Under this controlled scaling regime, **from-scratch multi-motion reinforcement learning exhibits earlier and more pronounced performance degradation as scale increases, whereas OMNIXTREME maintains higher tracking robustness over a broader scaling range.** As shown in Fig. 3, from-scratch multi-motion RL [28] exhibits a characteristic degradation pattern as motion diversity increases: tracking precision deteriorates steadily, followed by a sharp loss of robustness beyond a critical scale. These results indicate that the observed fidelity–scalability trade-off is not inherent, but can be substantially alleviated by a more scalable training paradigm.

E. Capacity scaling (Q3)

We next examine whether increasing model capacity further improves multi-motion tracking performance, and whether our generative policy exhibits stronger scaling behavior than conventional MLP-based controllers [33]. We train a family of models with increasing capacity (e.g., width/depth or Transformer hidden size and layers) under the same data and training recipe. Fig. 4 reports tracking fidelity and robustness as a function of model capacity. We observe that **additional capacity translates more directly into improved tracking quality for our flow matching policies, whereas MLP-based policies show weaker gains.**These results suggest that representational scaling is a practical lever for extending multi-motion tracking fidelity when paired with a scalable training paradigm.

F. Real-world executability and robustness (Q4)

We analyze the contribution of different post-training mechanisms to sim-to-real transfer by incrementally enabling them

并且随着动作多样性和难度的增加，采用保守的均值方法。

B. 评估指标

通过模拟运动跟踪的运行来评估策略，以提取性能指标。主要指标是成功率（Succ），如果人形机器人偏离参考运动的预定义阈值或变得不稳定，则该回合被视为不成功。我们额外报告了相对于根的每关节位置误差（MPJPE）（毫米），以及关节空间速度（Δvel）和加速度（Δacc）的差异，以量化运动学精度和物理保真度。

在物理机器人上，我们使用面向部署的指标来评估性能，包括技能级别的成功率和针对高动态行为的运动保真度的定性评估。

C. 可扩展的高保真跟踪（Q1）

在本节中，我们研究随着运动库的多样性和难度增加，OMNIXTREME是否仍能保持高保真人形机器人运动跟踪。我们在匹配模型容量和相同训练数据的情况下，将OMNIXTREME与专家到统一的MLP蒸馏以及从头开始的多运动强化学习进行比较。所有方法都在同一个组合运动库（LAFAN1 +XtremeMotion）上进行训练，并在三个测试集上进行评估：完整运动库、高动态XtremeMotion子集以及一个未见过的运动集（从重定向的AMASS中随机采样）。

仿真实验。如表II所示，OMNIXTREME在所有仿真指标上始终优于两种基线方法。在XtremeMotion和未见过的高动态运动中，差距变得更大，因为随着运动难度的增加，基线方法的成功率降低，跟踪误差增大。这表明OMNIXTREME在运动多样性和难度增加时仍能保持跟踪保真度，而不是在复杂性增加时性能下降。

真实世界。我们进一步在Unitree G1人形机器人上部署OMNIXTREME，使用来自XtremeMotion的运动。为了清晰展示，根据共享的动态结构和接触模式，将运动分组为代表不同技能类别的典型类别。如果运动无需人工干预或安全触发终止即可执行，则视为成功。如表III所示，在涵盖24种不同高动态运动的157次真实世界试验中，OMNIXTREME在包括翻转、杂技、霹雳舞和武术风格运动在内的各种技能类别中始终实现高成功率。这些结果表明，在仿真中观察到的可扩展性收益转化为在真实硬件上稳健且可物理执行的智能行为。

D. 保真度-可扩展性权衡 (Q2)

为表征多运动跟踪中的保真度-可扩展性权衡，我们通过在从XtremeMotion扩展的运动集中逐步训练来增加运动多样性，并分析不同训练范式在相同评估协议下的响应情况。

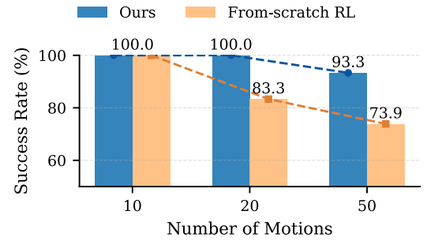


图3：保真度-可扩展性权衡。随着逐步增加运动多样性和难度，追踪成功率，同时评估所有策略在第一组10个运动上的表现。

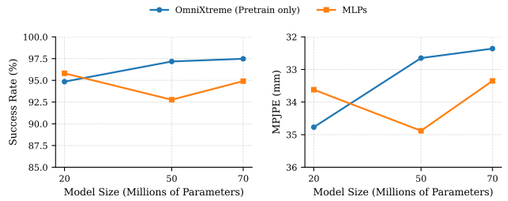


图4：容量扩展。随着模型容量的变化，追踪保真度和鲁棒性。OMNIXTREME从扩展中获益更大，而传统MLP控制器更早达到饱和。

在此受控扩展机制下，从头开始的多运动强化学习随着规模增加表现出更早且更明显的性能退化，而OMNIXTREME在更宽的可扩展范围内维持了更高的跟踪鲁棒性。如图3所示，从头开始的多运动RL [28] 随着运动多样性增加呈现出特征性退化模式：跟踪精度持续恶化，并在超过临界规模后出现鲁棒性的急剧损失。这些结果表明，观察到的保真度-可扩展性权衡并非固有属性，而是可以通过更具可扩展性的训练范式得到显著缓解。

E. 容量扩展 (Q3)

接下来，我们考察进一步增加模型容量是否能够提升多运动追踪性能，以及我们的生成策略是否比基于MLP的传统控制器表现出更强的扩展性 [33]。我们在相同的数据和训练方案下训练一系列具有不同容量的模型（例如，宽度/深度或Transformer隐藏大小和层数）。图4报告了模型容量与追踪保真度和鲁棒性的关系。我们观察到，对于我们的流匹配策略，额外容量能更直接地转化为追踪质量的提升，而基于MLP的策略收益较弱。这些结果表明，表示扩展在与可扩展性训练范式结合时，是一种实用的工具，可以扩展多运动追踪的保真度。

F. 实际可执行性与鲁棒性 (Q *4)*

我们通过逐步启用它们来分析不同后训练机制对仿真到现实迁移的贡献，并评估在技能层面的实际执行效果。表IV总结了消融实验结果。

TABLE II: **Scalable high-fidelity motion tracking under diverse motion sets.** OMNIXTREME consistently achieves lower kinematic errors and higher success rates than baselines, particularly on high-dynamic and unseen motions.

Method	LaFAN1+XtremeMotion				XtremeMotion				Unseen Motions	
	MPJPE ↓	Δvel ↓	Δacc ↓	Succ.(%) ↑	MPJPE ↓	Δvel ↓	Δacc ↓	Succ.(%) ↑	MPJPE ↓	Succ.(%) ↑
From-scratch RL [28]	47.95	10.03	3.27	82.95	54.19	14.04	4.04	79.45	56.87	85.29
Specialist→Unified MLP [58]	33.35	6.70	2.11	94.91	43.43	11.38	2.51	89.22	58.94	85.95
OmniXtreme (Pretrain only)	32.65	6.34	**2.04**	97.17	37.11	10.46	**2.39**	95.16	56.25	89.23
OmniXtreme (Pretrain + Post-train)	**30.93**	**6.19**	2.13	**98.54**	**36.17**	**9.94**	2.58	**95.64**	**56.05**	**89.54**

TABLE III: **Real-world evaluation of OMNIXTREME on Unitree G1.** We evaluate OMNIXTREME on physical hardware using motions drawn from the XtremeMotion motion library.

Skill	#Motions	Attempts	Success (%)↑
Flip	7	55	96.36
Handspring	5	35	88.57
Acrobatics	4	15	80.00
Breakdance	5	22	86.36
Martial arts	3	30	93.33
Total	**24**	**157**	**91.08**

TABLE IV: **Ablation of post-training mechanisms.** Real-world executability of different skills under incremental post-training mechanisms. **None**: base pretrained policy only; **MC**: motor constraints; **ADR**: aggressive domain randomization; **PS**: power-safety regularization (overcurrent / regenerative protection). ✓: stable execution; △: unstable or inconsistent execution; ×: consistent failure; ⊖: failures primarily associated with power-safety protection, such as overcurrent or excessive regenerative braking.

Skill	None	+MC	+MC+ADR	Full (+MC+ADR+PS)
Flip	△	✓	✓	✓
Breakdance	△	△	✓	✓
Acrobatics	×	△	⊖	✓

and evaluating real-world execution at the skill level. Tab. IV summarizes the ablation results.

In summary, **different classes of high-dynamic motions exhibit distinct failure modes, and each execution-oriented mechanism addresses a complementary aspect of real-world executability**. For highly impulsive motions such as flips, enforcing actuator torque-speed constraints is sufficient to enable stable execution, as respecting motor envelopes prevents immediate hardware-level violations. Contact-rich skills such as breakdance and acrobatic motions remain unstable under motor constraints alone, but benefit substantially from aggressive domain randomization, which improves robustness to timing-sensitive contact perturbations. Motions involving high-speed impact buffering, such as acrobatic landings, remain challenging even with aggressive domain randomization, power-safety regularization is critical for these skills, as it mitigates failures caused by excessive transient braking loads and unsafe energy absorption during high-impact contacts. Together, these results show that reliable real-world execution emerges from the combined effects of actuation-aware modeling, robustness-oriented randomization, and energy-aware safety constraints.

G. Qualitative results on extreme motions (Q5)

Finally, we provide qualitative evidence that OMNIXTREME can exhibit agile and versatile whole-body skills across distinct motion styles and contact patterns, beyond what is captured by scalar tracking metrics. We visualize representative rollouts spanning stylistic motions from XtremeMotion. Fig. 5 highlights that OMNIXTREME can track qualitatively different motions with coherent whole-body coordination, complement the quantitative metrics in Q1-Q4 and illustrate the breadth of behaviors enabled by scalable generative pretraining and actuation-aware refinement. Please refer to the *Supp.* for additional qualitative results, including video demonstrations.

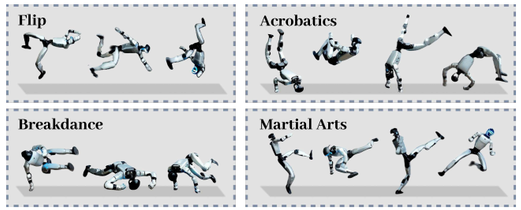


Fig. 5: **Qualitative results.** Representative real-world rollouts produced by OMNIXTREME, executing qualitatively distinct whole-body motions across diverse styles and contact patterns, including flips, acrobatics, breakdance, and martial-arts behaviors. The results illustrate stable and coordinated execution under rapid contact transitions and timing-sensitive phases on physical hardware.

V. CONCLUSION

We presented OMNIXTREME, a two-stage framework for scalable high-fidelity humanoid motion tracking in high-dynamic regimes. By combining specialist-to-unified flow-based pretraining with actuation-aware residual reinforcement learning, OMNIXTREME mitigates both the learning bottleneck at scale and the physical executability bottleneck at sim-to-real deployment. Extensive simulation results show that OMNIXTREME preserves tracking fidelity substantially deeper into motion diversity than other baselines, and real-robot experiments demonstrate reliable execution of diverse extreme behaviors with a single unified policy, breaking the conventional fidelity–scalability trade-off.

For future research, jointly scaling data diversity and model capacity will be essential for enhancing the generalization of whole-body humanoid motor skills. As learning-based controllers are pushed toward more dynamic and hardware-constrained regimes, actuation-aware modeling becomes a critical component of the learning pipeline. By incorporating high-fidelity actuation characteristics—such as current, power, torque, and speed-dependent constraints—researchers can further bridge the sim-to-real gap, ensuring that learned behaviors translate seamlessly to physical humanoid robots.

表 II：在不同运动集下的可扩展高保真运动跟踪。OMNIXTREME 持续比基线实现更低的运动学误差和更高的成功率，特别是在高动态和未见过运动上。

方法	LaFAN1+XtremeMotion				XtremeMotion				未见过运动	
	MPJPE ↓	Δ速度 ↓	Δ加速度 ↓	成功率(%) ↑	MPJPE ↓	Δ速度 ↓	Δ加速度 ↓	成功率(%) ↑	MPJPE ↓	成功率(%) ↑
从零开始的强化学习 [28]	47.95	10.03	3.27	82.95	54.19	14.04	4.04	79.45	56.87	85.29
专家→统一MLP [58]	33.35	6.70	2.11	94.91	43.43	11.38	2.51	89.22	58.94	85.95
OmniXtreme（仅预训练）	32.65	6.34	**2.04**	97.17	37.11	10.46	**2.39**	95.16	56.25	89.23
OmniXtreme（预训练 + 后训练）	**30.93**	**6.19**	2.13	**98.54**	**36.17**	**9.94**	2.58	**95.64**	**56.05**	**89.54**

表III：OmniXtreme在Unitree G1上的实际评估。我们使用来自XtremeMotion运动库的动作，在物理硬件上评估OmniXtreme。

技能	#动作	尝试	成功率 (%) ↑
Flip	7	55	96.36
手翻	5	35	88.57
杂技	4	15	80.00
霹雳舞	5	22	86.36
武术	3	30	93.33
总计	**24**	**157**	**91.08**

表IV：后训练机制的消融。增量后训练机制下不同技能的现实可执行性。None：仅基础预训练策略；MC：运动约束；ADR：激进领域随机化；PS：功率安全正则化（过流/再生保护）。✓：稳定执行；△：不稳定或不一致执行；×：一致失败；⊖：主要与功率安全保护相关的故障，如过流或再生制动过度。

技能	None	+MC	+MC+ADR	完整 (+MC+ADR+PS)
Flip	△	✓	✓	✓
霹雳舞	△	△	✓	✓
杂技	×	△	⊖	✓

和在实际技能层面的执行情况进行评估。表IV总结了消融实验结果。

总而言之，不同类型的高动态动作表现出不同的失效模式，而每种面向执行的机制都解决了现实世界可执行性中的一个互补方面。对于像翻转这样高度冲击性的动作，强制执行执行器扭矩-速度约束就足以实现稳定执行，因为遵守电机包络可以防止立即的硬件级违规。接触丰富的技能，如霹雳舞和杂技动作，仅靠电机约束仍然不稳定，但可以从激进的领域随机化中显著受益，这提高了对时间敏感的接触扰动的鲁棒性。涉及

高速冲击缓冲，例如空中动作落地，仍然具有挑战性，即使采用激进的领域随机化，功率安全正则化对这些技能至关重要，因为它减轻了由过度瞬态制动负荷和在高冲击接触期间不安全能量吸收引起的故障。这些结果共同表明，可靠的现实世界执行是由驱动感知建模、鲁棒性导向随机化和能量感知安全约束的综合效应产生的。

G. 极端动作的定性结果（Q5）

最后，我们提供定性证据，证明OMNIXTREME能够在不同场景中展现出敏捷且多变的全身技能

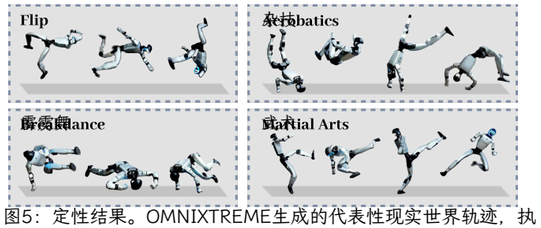


图5：定性结果。OMNIXTREME生成的代表性现实世界轨迹，执行不同风格和接触模式的定性不同全身动作，包括翻转、杂技、霹雳舞和武术行为。结果说明了在物理硬件上的快速接触转换和时间敏感阶段下的稳定和协调执行。

运动风格和接触模式，超越了标量跟踪指标所捕捉的范围。我们可视化了代表性的运行结果，涵盖了从XtremeMotion风格化的运动。图5突显了OMNIXTREME能够以协调的全身运动跟踪质地上不同的运动，补充了Q1-Q4中的定量指标，并说明了可扩展的生成式预训练和驱动感知改进所实现的行为广度。请参阅补充材料以获取额外的定性结果，包括视频演示。

V. 结论

我们提出了OMNIXTREME，这是一个用于高动态范围内可扩展高保真度人形运动跟踪的两阶段框架。通过结合专家到统一的基于流的预训练与驱动感知残差强化学习，OMNIXTREME缓解了规模学习的瓶颈和仿真到现实部署的物理可执行性瓶颈。广泛的仿真结果表明，OMNIXTREME比其他基线在运动多样性方面显著更深地保留了跟踪保真度，而真实机器人实验展示了使用单一统一策略可靠执行各种极端行为的性能，打破了传统的保真度-可扩展性权衡。

对于未来的研究，联合扩展数据多样性和模型容量对于提升全身人形机器人运动技能的泛化能力至关重要。随着基于学习的控制器被推向更动态和硬件受限的领域，驱动感知建模成为学习流程中的一个关键组成部分。通过整合高保真驱动特性——例如电流、功率、扭矩和与速度相关的约束——研究人员可以进一步缩小仿真到现实的差距，确保学习到的行为能够无缝地迁移到物理人形机器人上。

ACKNOWLEDGMENTS

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致谢

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APPENDIX

A. XtremeMotion Motion Library

The challenging motions selected from the AMASS dataset, Reallsusion, MimicKit, and LAFAN1 are detailed in Tab. V.

B. Motion Complexity Metrics

Given a motion sequence with frame index $t \in \{0, \dots, T-1\}$, frame time-step Δt, joint positions $q_t \in \mathbb{R}^J$, joint velocities $\dot{q}_t \in \mathbb{R}^J$, root/body positions $x_t \in \mathbb{R}^{B\times 3}$ and velocities $v_t \in \mathbb{R}^{B\times 3}$, we compute a set of scalar metrics that summarize its kinematic and contact complexity.

a) Joint/base kinematics.: From joint or base kinematics we extract:

$$v_{\max} = \max_{t,j} \left\| \dot{q}_t^{(j)} \right\|, \tag{7}$$

$$a_{\max} = \max_{t,j} \left\| \ddot{q}_t^{(j)} \right\|, \tag{8}$$

$$j_{\max} = \max_{t,j} \left\| \dddot{q}_t^{(j)} \right\|, \tag{9}$$

where $\ddot{q}_t$ and $\dddot{q}_t$ are obtained by finite differencing, e.g., $\ddot{q}_t \approx (\dot{q}_{t+1} - \dot{q}_t)/\Delta t$ and $\dddot{q}_t \approx (\ddot{q}_{t+1} - \ddot{q}_t)/\Delta t$. We use base (torso) linear and angular kinematics to define the same three scalars (maximum linear velocity, linear acceleration and angular velocity).

b) Center-of-mass vertical velocity.: Let z_t^{CoM} be the center-of-mass height and $\dot{z}_t^{\mathrm{CoM}}$ its vertical velocity. We measure the peak vertical CoM speed

$$v_{z,\max}^{\mathrm{CoM}} = \max_t \left| \dot{z}_t^{\mathrm{CoM}} \right|. \tag{10}$$

This captures the dynamic "liveliness" of the motion (e.g., jumps).

c) Airborne ratio.: Given a set of foot body indices $\mathcal{F}$, a frame is considered *airborne* if all feet are above a small height threshold h_{air} above the ground. The airborne ratio is the fraction of airborne frames

$$\text{Airborne} = \frac{1}{T} \sum_{t=0}^{T-1} \mathbf{1}\left[\min_{b\in\mathcal{F}} z_t^{(b)} > h_{\text{air}} \right], \tag{11}$$

where $z_t^{(b)}$ denotes the vertical position of body b at time t.

d) Contact switch frequency.: Independently, we obtain a binary contact state $c_t \in \{0,1\}^K$ for K end-effectors and define the contact switch frequency as the number of contact state flips per second:

$$f_{\text{switch}} = \frac{1}{(T-1)\,\Delta t} \sum_{t=0}^{T-2} \mathbf{1}[c_{t+1} \neq c_t] \ \ [\text{Hz}]. \tag{12}$$

In the radar plots, the *contact switch* axis corresponds to a normalized version of this scalar, $s_{\text{sw}} = \min(f_{\text{switch}}/10, 1)$.

e) Difficulty scores.: For comparability across metrics with different physical units we map the main scalar metrics to $[0,1]$ "difficulty" scores using simple clamping and linear scaling, mirroring the implementation used in our analysis code. Let

$$s_{ang} = \min\left(\frac{ang_{\max}}{20}, 1\right),$$
$$s_v = \min\left(\frac{v_{\max}}{20}, 1\right),$$
$$s_a = \min\left(\frac{a_{\max}}{200}, 1\right),$$
$$s_{\text{air}} = \text{clip}(\text{Airborne}, 0, 1),$$
$$s_{\text{com}} = \min\left(\frac{v_{z,\max}^{\mathrm{CoM}}}{2}, 1\right),$$
$$s_{\text{sw}} = \min\left(\frac{f_{\text{switch}}}{10}, 1\right),$$

where $\text{clip}(x,0,1) = \max(0, \min(x,1))$. The resulting 6-D score vector

$$\mathbf{s} = \left[s_{ang},\, s_v,\, s_a,\, s_{\text{com}},\, s_{\text{air}},\, s_{\text{sw}} \right] \tag{13}$$

is what we use in the radar plots to summarize and compare motion complexity across datasets.

C. Teacher Training

The teacher policy is trained using BeyondMimic [28]——a RL motion tracking algorithm within the IsaacLab simulator, with the specific reward constituents detailed in Tab. VII. Regarding the robot configuration, a full-mesh representation is employed for the collision geometry to ensure high-fidelity physical interactions. System parameters—including armature, k_p, k_d, and action scale—are kept consistent with the BeyondMimic framework, which is specifically tailored to the motor specifications of each joint.

The Teacher Policy employs an MLP architecture with hidden dimensions of [512, 256, 128] for both actor and critic. The actor observation is defined by

$$\mathbf{o} = [\boldsymbol{\psi},\, \mathbf{e}_{\text{torso}},\, \mathcal{V}_{\text{imu}},\, \boldsymbol{q} - \boldsymbol{q}^0,\, \dot{\boldsymbol{q}},\, \mathbf{a}_{\text{last}}],$$

including reference motion joint positions and velocities $\boldsymbol{\psi} = [\boldsymbol{q}^{\text{ref}}, \dot{\boldsymbol{q}}^{\text{ref}}]$, $\mathbf{e}_{\text{torso}}$ which includes torso position difference and torso orientation difference from reference represented by the 6D orientation from $R_{\text{torso}}^{\text{ref}} R_{\text{torso}}^{\top}$, base linear and angular velocities $\mathcal{V}_{\text{imu}} \in \mathbb{R}^6$, relative joint positions $\boldsymbol{q} - \boldsymbol{q}^0$ computed by joint position $\boldsymbol{q}$ subtract default joint position $\boldsymbol{q}^0$ and joint velocities $\dot{\boldsymbol{q}}$, and the previous action $\mathbf{a}_{\text{last}}$, all of which are subject to noise described in Tab. I to enhance robustness. In contrast, the critic observation includes these same elements in their ground-truth form, supplemented by privileged information such as the robot's precise body position and orientation, providing a comprehensive and noise-free state representation to facilitate stable value function estimation during training.

The detailed training parameters are shown in Tab. VI. To speed up training, early termination is introduced when tracking errors are larger than a threshold. Tracking errors are defined as $\mathbf{e}_{p,b} = \mathbf{p}_b^{\text{ref}} - \mathbf{p}_b$ for position and $\mathbf{e}_{R,b} = \log(R_b^{\text{ref}} R_b^{\top})$ for

附录

A. XtremeMotion 运动库

从 AMASS 数据集、Reallusion、MimicKit 和 LAFAN1 中选择的具有挑战性的动作在表 V 中详细说明。

B. 运动复杂度指标

给定一个具有帧索引 $t \in \{0, \dots, T-1\}$ 的运动序列、帧时间步长 Δt、关节位置 $q_t \in \mathbb{R}^J$、关节速度 $\dot{q}_t \in \mathbb{R}^J$、根/身体位置 $x_t \in \mathbb{R}^{B\times 3}$ 和速度 $v_t \in \mathbb{R}^{B\times 3}$，我们计算一组标量指标，以总结其运动学和接触复杂度。

a) 关节/基座运动学：从关节或基座运动学中提取：

$$v_{\max} = \max_{t,j} \left\| \dot{q}_t^{(j)} \right\|, \tag{7}$$

$$a_{\max} = \max_{t,j} \left\| \ddot{q}_t^{(j)} \right\|, \tag{8}$$

$$j_{\max} = \max_{t,j} \left\| \dddot{q}_t^{(j)} \right\|, \tag{9}$$

其中 $\ddot{q}_t$ 和 … q_t 通过有限差分获得，例如，$\ddot{q}_t \approx (\dot{q}_{t+1} - \dot{q}_t)/\Delta t$ 和 … $q_t \approx (\ddot{q}_{t+1} - \ddot{q}_t)/\Delta t$。我们使用基座（躯干）的线性和角运动学来定义相同的三个标量（最大线速度、线加速度和角速度）。

b) 质心垂直速度：设 z_t^{CoM}为质心高度， $\dot{z}_t^{\mathrm{CoM}}$为其垂直速度。我们测量质心垂直速度的峰值

$$v_{z,\max}^{\mathrm{CoM}} = \max_t \left| \dot{z}_t^{\mathrm{CoM}} \right|. \tag{10}$$

这捕捉了运动的动态"活力"（例如跳跃）

c) 空中比率：给定一组足部身体索引 $\mathcal{F}$，如果所有足部都高于地面一个小的阈值 h_{air}，则该帧被视为空中帧。空中比率是空中帧的比例

$$\text{Airborne} = \frac{1}{T} \sum_{t=0}^{T-1} \mathbf{1}\left[\min_{b\in\mathcal{F}} z_t^{(b)} > h_{\text{air}} \right], \tag{11}$$

其中 $z_t^{(b)}$ 表示身体 b在时间 t的垂直位置

d) 接触切换频率：独立地，我们为 K 末端执行器获得一个二进制接触状态 $c_t \in \{0,1\}^K$，并将接触切换频率定义为每秒接触状态翻转的次数：

$$f_{\text{switch}} = \frac{1}{(T-1)\,\Delta t} \sum_{t=0}^{T-2} \mathbf{1}[c_{t+1} \neq c_t] \ \ [\text{Hz}]. \tag{12}$$

在雷达图上，接触开关轴对应于该标量的归一化版本，$s_{\text{sw}} = \min(f_{\text{switch}}/10, 1)$。

e) 难度分数：为了在不同物理单位的不同指标之间进行比较，我们将主要标量指标映射到 $[0,1]$ "难度"分数，使用简单的夹断和线性缩放，与我们分析代码中使用的实现方式相同。让

$$s_{ang} = \min\left(\frac{ang_{\max}}{20}, 1\right),$$
$$s_v = \min\left(\frac{v_{\max}}{20}, 1\right),$$
$$s_a = \min\left(\frac{a_{\max}}{200}, 1\right),$$
$$s_{\text{air}} = \text{clip}(\text{Airborne}, 0, 1),$$
$$s_{\text{com}} = \min\left(\frac{v_{z,\max}^{\mathrm{CoM}}}{2}, 1\right),$$
$$s_{\text{sw}} = \min\left(\frac{f_{\text{switch}}}{10}, 1\right),$$

其中$\text{clip}(x,\ 0,\ 1)\ = \max(0,\ \min(x,\ 1))$。得到的6-D评分向量

$$\mathbf{s} = \left[s_{ang},\, s_v,\, s_a,\, s_{\text{com}},\, s_{\text{air}},\, s_{\text{sw}} \right] \tag{13}$$

这是我们在雷达图上用来总结和比较跨数据集运动复杂性的方法。

C. 教师训练

教师策略使用 BeyondMimic [28] 进行训练——这是一种在 IsaacLab 模拟器中的强化学习运动跟踪算法，其具体的奖励组成部分详见表 VII。关于机器人配置，碰撞几何采用全网格表示以确保高保真物理交互。系统参数——包括骨架、k_p、 k_d 和动作缩放——与 BeyondMimic 框架保持一致，该框架专门针对每个关节的电机规格进行了定制。

教师策略采用具有 [512, 256, 128] 隐藏维度的 MLP 架构，用于演员和评价者。演员的观察定义为

$$\mathbf{o} = [\boldsymbol{\psi},\, \mathbf{e}_{\text{torso}},\, \mathcal{V}_{\text{imu}},\, \boldsymbol{q} - \boldsymbol{q}^0,\, \dot{\boldsymbol{q}},\, \mathbf{a}_{\text{last}}],$$

包括参考运动关节位置和速度 $\boldsymbol{\psi} = [\boldsymbol{q}^{\text{ref}}, \dot{\boldsymbol{q}}^{\text{ref}}]$, $\mathbf{e}_{\text{torso}}$ （其中包含躯干位置差和躯干方向差，由 $R_{\text{torso}}^{\text{ref}} R_{\text{torso}}^{\top}$ 表示的 6D 方向表示），基座线速度和角速度 $\mathcal{V}_{\text{imu}} \in \mathbb{R}^6$，相对关节位置 $\boldsymbol{q} - \boldsymbol{q}^0$ （通过关节位置 $\boldsymbol{q}$ 减去默认关节位置 $\boldsymbol{q}^0$ 计算得到）和关节速度 $\dot{\boldsymbol{q}}$，以及先前的动作 a_{last}，所有这些都受到表 I 中描述的噪声影响以增强鲁棒性。相比之下，评价者的观察包括这些相同元素的真实形式，并辅以特权信息，例如机器人的精确身体位置和方向，提供全面且无噪声的状态表示，以促进训练期间稳定值函数的估计。

详细的训练参数显示在表VI中。为了加快训练速度，当跟踪误差大于阈值时引入提前终止。跟踪误差定义为位置上的 $e_{p,b} = \mathrm{p}_b^{\text{ref}} - \mathrm{p}_b$， 以及 $e_{R,b} = \log(R_b^{\text{ref}} R_b^{\top})$。

TABLE V: Motion information summary.

Motion ID	Motion Source	Motion Description
1	CMU-85-05	Handstand walk.
2	CMU-85-10	Handstand spin.
3	CMU-88-09	Back Handspring with a full twist.
4	CMU-88-08	Back Handspring with a half twist.
5	CMU-90-08	Aerial Cartwheel
6	CMU-85-14	B-boying with a rapid-fire string of back handsprings.
7	CMU-85-13	Skip back, pivot through a hand-to-headstand, drop, flip, and bounce back.
8	CMU-90-06	Fly kick.
9	CMU-90-34	Forward roll.
10	CMU-90-01	Backward roll.
11	CMU-90-28	Backspin.
12	CMU-85-08	Thomas Flare.
13	CMU-85-12	Long breaking dance.
14	CMU-85-04	Another long breaking dance.
15	CMU-85-01	Bicycle kick flip.
16	CMU-85-02	Another bicycle kick flip.
17	CMU-88-06	Butterfly kick.
18	CMU-85-06	Webster flip.
19	CMU-49-08	Two consecutive cartwheels.
20	CMU-90-30	Alternating pistol squats.
21	CMU-90-29	Acrobatic gymnastics with cartwheel and back handsprings.
22	CMU-90-19	Crawl forward and backflip.
23	GeneralA11-MilitaryCrawlForward	Stay low and crawl forward.
24	IconicHeroMotion-SwordJudgment	Spinning slash.
25	IconicHeroMotion-SwordHeroic	Another style of spinning slash.
26	HandtoHandCombat-B3AttackReverseTurningKick	Reverse turning kick.
27	HandtoHandCombat-D2AttackPunchSweepKick	Punch and sweep kick.
28	HandtoHandCombat-D4AttackReverseFrontSnapKick	Reverse and front snap kick.
29	HandtoHandCombat-D4DodgeRollBack	Two consecutive rolls in different styles.
30	HandtoHandCombat-G1GetupKipUp	Kip up.
31	HandtoHandCombat-G2GetupHandstandKipUp	Handstand kip.
32	HandtoHandCombat-KO2FalltoGroundAxelDown	Execute a downward Axel into a ground fall.
33	LadyAgent-AgentElbowStrikeSweepKick	Elbow strike and sweep kick.
34	LadyAgent-AgentHandspring	Front handspring.
35	LadyAgent-AgentRollForward	Shoulder roll.
36	LadyAgent-AgentShootSidewardRoll	Quick side roll.
37	LadyAgent-AgentSnapKick	Snapkick.
38	Mimickit-g1spinkick	Spinkick.
39	LAFAN1-dance1subject2 [82.8,106.9]s	Constantly spin in full circles.
40	LAFAN1-dance1subject2 [145.3,161.3]s	Play guitar while hopping on one leg.
41	LAFAN1-dance2subject3 [160.2,224.3]s	Flutter arms and hands.
42	LAFAN1-fightandsports1subject1 [167.0,176.9]s	Continuous long jumps.
43	LAFAN1-fightandsports1subject1 [0.0,17.0]s	Balance on one leg.
44	LAFAN1-dance1subject1 [104.6,119.1]s	Cartwheel twice.
45	LAFAN1-jump1subject1 [70.3,87.6]s	Play hopscotch.
46	LAFAN1-jump1subject1 [89.1,138.4]s	Hop on one foot.
47	LAFAN1-jump1subject1 [0.0,72.0]s	Successive leaps.
48	LAFAN1-fightandsports1subject4 [154.4,219.8]s	Vigorously swing the baseball bat.
49	LAFAN1-fightandsports1subject4 [61.1,85.9]s	Lash the golf club.
50	LAFAN1-fightandsports1subject4 [28.2,61.9]s	Diverse kicking movements.
51	LAFAN1-run1subject2 [20.4,101.0]s	Shuttle run.
52	LAFAN1-run1subject2 [92.0,130.6]s	Run rapidly.
53	LAFAN1-fallandgetup2subject3 [33.8,56.1]s	Kip up twice.
54	LAFAN1-fightandsports1subject1 [17.3,27.2]s	Roundhouse kick.
55	LAFAN1-jump1subject2 [187.9,196.2]s	Push up.
56	LAFAN1-jump1subject2 [196.0,205.8]s	Lateral roll.
57	LAFAN1-jump1subject2 [205.6,244.4]s	Lateral roll and kip up.

表 V：运动信息摘要。

运动 ID	运动来源	运动描述
1	CMU-85-05	倒立行走。
2	CMU-85-10	倒立旋转。
3	CMU-88-09	后空翻带全旋。
4	CMU-88-08	后空翻带半旋。
5	CMU-90-08	空中翻轮
6	CMU-85-14	快速连续后手翻的霹雳舞动作。
7	CMU-85-13	向后跳跃，手转成头倒立，落地，翻转，再弹回。
8	CMU-90-06	飞踢。
9	CMU-90-34	前滚翻。
10	CMU-90-01	后滚翻。
11	CMU-90-28	反旋球。
12	CMU-85-08	托马斯旋转。
13	CMU-85-12	长板街舞。
14	CMU-85-04	另一段长时间的霹雳舞。
15	CMU-85-01	自行车踢翻。
16	CMU-85-02	另一段自行车踢翻。
17	CMU-88-06	蝴蝶踢
18	CMU-85-06	韦伯翻转
19	CMU-49-08	连续两个翻滚
20	CMU-90-30	交替深蹲.
21	CMU-90-29	杂技体操，包含翻滚和后空翻.
22	CMU-90-19	向前爬行和后空翻.
23	GeneralA11-军事匍匐前进	保持低姿态，匍匐前进。
24	IconicHeroMotion-剑裁决	旋转斩击。
25	IconicHeroMotion-剑英勇	另一种旋转斩击风格。
26	近身格斗-B3攻击反向转身踢	反向转身踢。
27	近身格斗-D2攻击拳击扫踢	拳击和扫踢。
28	近身格斗-D4攻击反向前踢	反向和前踢。
29	近身格斗-D4后翻	连续使用不同风格的翻滚动作。
30	近身格斗-G1起身-翻滚起身	翻滚起身。
31	近身格斗-G2起身-手倒立翻滚	手倒立翻滚。
32	近身格斗-倒地阿克塞尔下劈	执行阿克塞尔下劈至倒地动作。
33	女特工-肘击扫踢	肘击与扫踢。
34	女特工-前空翻	前空翻。
35	LadyAgent-Agent向前翻滚	肩部翻滚。
36	LadyAgent-Agent侧向翻滚射击	快速侧向翻滚。
37	LadyAgent-Agent迅猛踢	迅猛踢。
38	Mimickit-g1spinkick	Spinkick.
39	LAFAN1-舞蹈1主题2 [82.8,106.9]s	不断原地转圈。
40	LAFAN1-舞蹈1主题2 [145.3,161.3]s	单腿蹦跳时弹吉他。
41	LAFAN1-舞蹈2主题3 [160.2,224.3]s	挥动手臂和双手。
42	LAFAN1-格斗与运动1主题1 [167.0,176.9]s	连续长跳。
43	LAFAN1-格斗与运动1主题1 [0.0,17.0]s	单腿平衡。
44	LAFAN1-舞蹈1主题1 [104.6,119.1]s	翻滚两次。
45	LAFAN1-跳跃1主题1 [70.3,87.6]s	玩跳房子。
46	LAFAN1-跳跃1主题1 [89.1,138.4]s	单脚跳。
47	LAFAN1跳跃1主题1 [0.0,72.0]s	连续跳跃。
48	LAFAN1战斗与运动1主题4 [154.4,219.8]s	用力挥动棒球棒。
49	LAFAN1战斗与运动1主题4 [61.1,85.9]s	挥动高尔夫球杆。
50	LAFAN1-格斗与运动1主题4 [28.2,61.9]s	多种踢腿动作。
51	LAFAN1-跑步1主题2 [20.4,101.0]s	折返跑。
52	LAFAN1-跑步1主题2 [92.0,130.6]s	快速奔跑。
53	LAFAN1-跌起2主题3 [33.8,56.1]s	连续抬起两次。
54	LAFAN1-战斗与运动1主题1 [17.3,27.2]s	旋风踢。
55	LAFAN1跳1主题2 [187.9,196.2]s	俯卧撑。
56	LAFAN1跳跃1主题2 [196.0,205.8]s	侧向滚动。
57	LAFAN1跳跃1主题2 [205.6,244.4]s	侧向滚动和向上踢。

orientation. Termination is triggered if the vertical position error $|\mathbf{e}_{p,z,b}|$ of the torso or any end-effector ($\mathcal{B}_{\text{ee}}$) exceeds 0.25m, or if the local gravity vector discrepancy caused by torso's orientation error $\|\mathbf{e}_{R,\text{torso}}\|$ exceeds 0.8.

TABLE VI: **Motion tracking hyperparameters for teacher training** [28].

Hyperparameter	Value
Architecture	
Actor MLP hidden dimensions	[512, 256, 128]
Critic MLP hidden dimensions	[512, 256, 128]
Activation function	ELU
Training	
Steps per environment	24
Clip parameter	0.2
Entropy coefficient	0.005
Value loss coefficient	1.0
Discount factor (γ)	0.99
GAE λ	0.95
Desired KL	0.01
Learning epochs	5
Mini-batches	4

D. Flow matching policy training

The inputs robot proprioception p and motion command c are each mapped to a single token, while the 15-step history h is mapped to 15 individual tokens, resulting in a concatenated sequence of 17 tokens. Each mapping is performed by a dedicated MLP preprocessing unit consisting of two 256-unit layers with ReLU activations. This sequence is then processed by a Transformer encoder, followed by global average pooling and a linear projection to produce a 1024-dimensional embedding.

Simultaneously, the action a_t is projected through a linear layer to a 1024-dimensional space, while the time step t is transformed into a 1024-dimensional embedding using sinusoidal encoding. These three components—the state embedding, action embedding, and time embedding—are concatenated and fed into a deep MLP with hidden dimensions of $[2048, 2048, 2048]$ to predict the velocity v_t at time t.

E. Residual policy training

The Residual Policy is implemented as an MLP with hidden layer sizes of [1024, 512, 256, 128, 64], with the corresponding reward functions detailed in Tab. VIII.

The residual learning framework employs an asymmetric actor-critic architecture, as well. In this setup, the residual actor's observation space is restricted to proprioceptive data, relative motion torso orientation, motion command joint pos and joint velocity, the previous step's total action, and the base action out by the flow matching policy. To enhance sim-to-real robustness, uniform noise is injected into these observations, and the noise is aligned with the flow matching base policy noise.

Conversely, the residual critic utilizes privileged information available only during simulation. The critic's observation space encompasses the full ground-truth state, including noise-free versions of the actor's inputs, as well as critical state variables hidden from the actor, such as the robot's global body position and orientation, base linear velocity, and the precise position of the motion torso.

TABLE VII: Reward function terms and expressions for Teacher Policy training [28].

Reward Term	**Weight**	**Expression**
Global Torso Position	0.5	$\exp(-\|p_{err}\|^2/0.3^2)$
Global Torso Orientation	0.5	$\exp(-\|\theta_{err}\|^2/0.4^2)$
Relative Body Position	1.0	$\exp(-\|p_{rel_err}\|^2/0.3^2)$
Relative Body Orientation	1.0	$\exp(-\|\theta_{rel_err}\|^2/0.4^2)$
Body Linear Velocity	1.0	$\exp(-\|v_{err}\|^2/1.0^2)$
Body Angular Velocity	1.0	$\exp(-\|\omega_{err}\|^2/3.14^2)$
Action Rate	-0.1	$\|a_t - a_{t-1}\|^2$
Joint Limit	-10.0	$\sum \max(0, q - q_{limit})$
Undesired Contacts	-0.1	$\sum 1(F_{contact} > 1.0)$

TABLE VIII: Reward function terms and expressions for Residual Policy training. Refer to Sec. V-K for further details regarding power safety regularization.

Reward Term	**Weight**	**Expression**
Global Torso Position	0.5	$\exp(-\|p_{err}\|^2/0.3^2)$
Global Torso Orientation	0.5	$\exp(-\|(\theta_{err}\|^2/0.4^2)$
Relative Body Position	1.0	$\exp(-\|p_{rel_err}\|^2/0.3^2)$
Relative Body Orientation	1.0	$\exp(-\|\theta_{rel_err}\|^2/0.4^2)$
Body Linear Velocity	1.0	$\exp(-\|v_{err}\|^2/1.0^2)$
Body Angular Velocity	1.0	$\exp(-\|\omega_{err}\|^2/3.14^2)$
Action Rate	-0.1	$\|a_t - a_{t-1}\|^2$
Joint Limit	-10.0	$\sum \max(0, q - q_{limit})$
Undesired Contacts	-0.1	$\sum 1(F_{contact} > 1.0)$
Power Safety Regularization	-10.0	$\sum \left(\frac{\max(0,\ -\tau\dot{q}-150)}{500}\right)^2$

F. Actuator Modeling

The physical parameters of the actuators, including armature inertia I, torque-speed characteristics, and friction coefficients, are summarized in Tab. IX.

TABLE IX: **Actuator modeling parameters.**

Actuator	τ_{y1}	τ_{y2}	v_{x1}	v_{x2}	μ_s	v_{act}	μ_d	I
5020-16	24.8	31.9	30.86	40.13	0.6	0.01	0.06	3.610e-03
7520-14.3	71.0	83.3	22.63	35.52	1.6	0.01	0.16	1.018e-02
7520-22.5	111.0	131.0	14.5	22.7	2.4	0.01	0.24	2.510e-02
4010-25	4.8	8.6	15.3	24.76	0.6	0.01	0.06	4.250e-03

G. PD control

Following the methodology of BeyondMimic, we calculate the PD gains, k_p and k_d, based on the natural frequency (ω), damping ratio (ζ), and the armature inertia (I) of the motors:

$$k_{\text{p},j} = I_j\omega^2, \quad k_{\text{d},j} = 2I_j\zeta\omega$$

For the configuration addressed here, the parameters consist of a 10 Hz natural frequency and a damping ratio of 2. To determine the action scale, we still utilize the maximum torque

方向。如果躯干的垂直位置误差 $|e_{p,z,b}|$ 或任何末端执行器 ($\mathcal{B}_{\text{ee}}$) 超过 0.25 米，或者躯干方向误差 $\|e_{R,\text{torso}}\|$ 导致的局部重力矢量偏差超过 0.8，则触发终止。

表 ***VI***：教师训练 [28] 的运动跟踪超参数。

超参数	值
架构	
Actor MLP 隐藏维度	[512, 256, 128]
Critic MLP 隐藏维度	[512, 256, 128]
激活函数	ELU
训练	
每个环境的步数	24
裁剪参数	0.2
熵系数	0.005
值损失系数	1.0
折扣因子 (γ)	0.99
GAE λ	0.95
期望的 KL	0.01
学习轮数	5
小批量	4

D. 流匹配策略训练

机器人的本体感觉 p 和运动指令 c 各映射为一个单独的 token，而 15 步历史 h 则映射为 15 个独立的 token，最终形成由 17 个 token 组成的串联序列。每个映射由一个专用的 MLP 预处理单元执行，该单元包含两层 256 个单元的 ReLU 激活层。该序列随后被 Transformer 编码器处理，接着进行全局平均池化，并通过线性投影生成一个 1024 维的嵌入。

同时，动作 a_t 通过线性层投影到 1024 维空间，而时间步 t 则使用正弦编码转换为 1024 维的嵌入。这三个组件——状态嵌入、动作嵌入和时间嵌入——被串联并输入到一个隐藏维度为 $[2048, 2048, 2048]$ 的深度 MLP 中，以预测时间 t 的速度 v_t。

E. 残差策略训练

残差策略作为MLP实现，其隐藏层大小为 [1024, 512, 256, 128, 64], ，相应的奖励函数详见表VIII。

残差学习框架还采用了一种非对称的Actor-Critic架构。在此设置中，残差Actor的观测空间仅限于本体感觉数据、相对运动躯干方向、运动指令关节位置和关节速度、前一步的总动作以及流匹配策略输出的基础动作。为增强仿真到现实的鲁棒性，将均匀噪声注入这些观测数据中，且噪声与流匹配基础策略的噪声对齐。

相反，残差评价者利用的是仅在模拟过程中可获得的特权信息。评价者的观测空间包含完整的真实状态，包括无噪声的

表VII：教师策略训练的奖励函数项和表达式 [28]。

奖励项	权重	表达式
全局躯干位置	0.5	$\exp(-\|p_{err}\|^2/0.3^2)$
全局躯干朝向	0.5	$\exp(-\|\theta_{err}\|^2/0.4^2)$
相对身体位置	1.0	$\exp(-\|p_{rel_err}\|^2/0.3^2)$
相对身体朝向	1.0	$\exp(-\|\theta_{rel_err}\|^2/0.4^2)$
身体线速度	1.0	$\exp(-\|v_{err}\|^2/1.0^2)$
身体角速度	1.0	$\exp(-\|\omega_{err}\|^2/3.14^2)$
动作率	-0.1	$\|a_t - a_{t-1}\|^2$
关节限制	-10.0	$\sum \max(0, q - q_{limit})$
非期望接触	-0.1	$\sum 1(F_{contact} > 1.0)$

表 VIII：残差策略训练的奖励函数项和表达式。有关功率安全正则化的更多详细信息，请参阅第 V-K 节。

奖励项	权重	表达式
全局躯干位置	0.5	$\exp(-\|p_{err}\|^2/0.3^2)$
全局躯干朝向	0.5	$\exp(-\|(\theta_{err}\|^2/0.4^2)$
相对身体位置	1.0	$\exp(-\|p_{rel_err}\|^2/0.3^2)$
相对身体朝向	1.0	$\exp(-\|\theta_{rel_err}\|^2/0.4^2)$
身体线速度	1.0	$\exp(-\|v_{err}\|^2/1.0^2)$
身体角速度	1.0	$\exp(-\|\omega_{err}\|^2/3.14^2)$
动作率	-0.1	$\|a_t - a_{t-1}\|^2$
关节限制	-10.0	$\sum \max(0, q - q_{limit})$
不期望的接触	-0.1	$\sum 1(F_{contact} > 1.0)$
功率安全正则化	-10.0	$\sum \left(\frac{\max(0,\ -\tau\dot{q}-150)}{500}\right)^2$

版本，以及对演员隐藏的关键状态变量，例如机器人的全局身体位置和方向、基础线性速度以及运动躯干的精确位置。

F. 伺服器建模

伺服器的物理参数，包括电枢惯性 I、扭矩-速度特性以及摩擦系数，汇总于表IX。

表***IX***：执行器建模参数。

执行器	τ_{y1}	τ_{y2}	v_{x1}	v_{x2}	μ_s	v_{act}	μ_d	I
5020-16	24.8	31.9	30.86	40.13	0.6	0.01	0.06	3.610e-03
7520-14.3	71.0	83.3	22.63	35.52	1.6	0.01	0.16	1.018e-02
7520-22.5	111.0	131.0	14.5	22.7	2.4	0.01	0.24	2.510e-02
4010-25	4.8	8.6	15.3	24.76	0.6	0.01	0.06	4.250e-03

G. PD控制

遵循BeyondMimic的方法，我们根据电机的固有频率(ω)、阻尼比(ζ)和电枢惯性(I)计算PD增益， k_p 和 k_d:

$$k_{\text{p},j} = I_j\omega^2, \quad k_{\text{d},j} = 2I_j\zeta\omega$$

对于此处所讨论的配置，参数包括10 Hz的固有频率和阻尼比2。为确定动作尺度，我们仍然使用最大扭矩

TABLE X: Unitree G1 Joint-to-Motor Mapping.

Joint Name	Motor Model
Hip Pitch Joint	7520-22.5
Hip Roll Joint	7520-22.5
Knee Joint	7520-22.5
Hip Yaw Joint	7520-14.3
Ankle Pitch Joint	5020
Ankle Roll Joint	5020
Waist Roll Joint	5020
Waist Pitch Joint	5020
Waist Yaw Joint	7520-14.3
Shoulder Pitch Joint	5020
Shoulder Roll Joint	5020
Shoulder Yaw Joint	5020
Elbow Joint	5020
Wrist Roll Joint	5020
Wrist Pitch Joint	4010
Wrist Yaw Joint	4010

values ($\boldsymbol{\tau}_{\max}$) originally defined in the URDF, scaled by a factor of 0.25[28]:

$$\boldsymbol{\alpha} = 0.25 \frac{\boldsymbol{\tau}_{\max}}{k_{\mathrm{p},j}}$$

Notably, due to differences in the G1 robot variants, we utilize 7520-22.5 motors for the hip pitch actuators(Tab. X), rather than the 7520-14.3 models used in the original BeyondMimic.

The PD setpoints (target positions) are computed using the following relationship:

$$\boldsymbol{q}^{\mathrm{tar}} = \boldsymbol{q}^{0} + \boldsymbol{\alpha} \odot \mathbf{a}$$

where $\boldsymbol{q}^{0}$ represents the default joint positions, and $\mathbf{a}$ is the action output by the policy. The pre-clipped control torque is then calculated using the standard PD control law:

$$\boldsymbol{\tau} = k_p(\boldsymbol{q}^{\mathrm{tar}} - \boldsymbol{q}) - k_d\dot{\boldsymbol{q}}$$

In this expression, $\mathbf{q}$ and $\dot{\mathbf{q}}$ denote the current joint positions and velocities, respectively.

H. Evaluation Protocol and Metrics

This subsection details the evaluation protocol and metrics used throughout the paper. Our goal is to assess tracking fidelity, robustness, and generalization under controlled conditions that are consistent with training and reflective of sim-to-real performance.

a) Motion segmentation and episode length.: To avoid bias introduced by varying motion durations, all reference motions are segmented into fixed-length clips of **10 seconds** (corresponding to **500 control steps**), which matches the episode length used during training. Motions shorter than 10 seconds are evaluated without further segmentation. All reported metrics in simulation and real world are computed at the clip level.

b) Simulation evaluation settings.: During simulation evaluation, we retain the same sensor noise and base domain randomization used during training, including center-of-mass offsets and default joint position offsets. This ensures that evaluation reflects basic sim-to-real robustness rather than idealized noise-free tracking. Each motion clip is evaluated over **10 independent rollouts**, and metrics are averaged across these trials.

c) Unseen motion set.: In addition to test motions drawn from the training distributions, we evaluate generalization on an unseen motion set. This set is constructed by uniformly sampling **1000** motion clips (10 seconds each) from retargeted AMASS CMU and KIT mocap sequences, explicitly excluding all motions used in training or selected as extreme motions. The unseen set primarily consists of locomotion, turning, and simple dance-like behaviors, and is used to assess generalization beyond the curated extreme motion library.

d) Evaluation metrics.: We employ a set of pose-based and physics-aware metrics commonly used in humanoid motion tracking. All metrics are computed per episode and then averaged across episodes.

Let $t = 1, \ldots, T$ index control steps in an episode with step time Δt (seconds), and let $i = 1, \ldots, N$ index tracked bodies. All per-body errors are computed over the same tracked-body subset.

e) Root-relative body positions.: Reference motions are aligned to the robot using the torso pose. We denote the aligned reference body position as $\mathbf{p}_{t,i}^{\mathrm{ref}}$ and the robot body position as $\mathbf{p}_{t,i}^{\mathrm{rob}}$.

f) MPJPE (mm).:

$$\mathrm{MPJPE} = 1000 \cdot \frac{1}{T}\sum_{t=1}^{T}\left(\frac{1}{N}\sum_{i=1}^{N}\left\|\mathbf{p}_{t,i}^{\mathrm{ref}} - \mathbf{p}_{t,i}^{\mathrm{rob}}\right\|_2\right). \quad (14)$$

g) Δvel (mm/frame).: Let $\mathbf{v}_{t,i}^{\mathrm{ref}}$ and $\mathbf{v}_{t,i}^{\mathrm{rob}}$ denote linear velocities (in m/s).

$$\Delta v = 1000 \cdot \Delta t \cdot \frac{1}{T}\sum_{t=1}^{T}\left(\frac{1}{N}\sum_{i=1}^{N}\left\|\mathbf{v}_{t,i}^{\mathrm{ref}} - \mathbf{v}_{t,i}^{\mathrm{rob}}\right\|_2\right). \quad (15)$$

h) Δacc (mm/frame^2).: Acceleration is computed by finite differences on velocity:

$$\mathbf{a}_{t,i} = \frac{\mathbf{v}_{t,i} - \mathbf{v}_{t-1,i}}{\Delta t}.$$

The per-step acceleration error is:

$$e_t^{\mathrm{acc}} = \frac{1}{N}\sum_{i=1}^{N}\left\|\mathbf{a}_{t,i}^{\mathrm{ref}} - \mathbf{a}_{t,i}^{\mathrm{rob}}\right\|_2. \quad (16)$$

We exclude the step immediately following an environment reset. The reported acceleration error is:

$$\Delta a = 1000 \cdot \Delta t^2 \cdot \frac{1}{T}\sum_{t=1}^{T} e_t^{\mathrm{acc}}. \quad (17)$$

i) Success rate.: An episode is considered successful if it terminates due to time-out (i.e., runs for its full allotted duration) rather than early termination. **The termination criteria used during evaluation are identical to those used during training**, including violation of safety thresholds. Success rate is defined as the fraction of successful episodes.

表X：Unitree G1关节到电机映射。

关节名称	电机型号
髋关节俯仰	7520-22.5
髋关节偏航	7520-22.5
膝关节	7520-22.5
髋关节偏航	7520-14.3
踝关节俯仰	5020
踝关节侧向运动	5020
腰部侧向运动	5020
腰部俯仰运动	5020
腰部偏航运动	7520-14.3
肩部俯仰运动	5020
肩部滚动关节	5020
肩部偏航关节	5020
肘部关节	5020
手腕滚动关节	5020
手腕俯仰关节	4010
手腕偏航关节	4010

值（$\boldsymbol{\tau}_{\max}$）最初在URDF中定义，按0.25[28]的比例缩放：

$$\boldsymbol{\alpha} = 0.25 \frac{\boldsymbol{\tau}_{\max}}{k_{\mathrm{p},j}}$$

值得注意的是，由于G1机器人型号的差异，我们为髋关节俯仰执行器使用7520-22.5型号的电机（见表X），而非原始BeyondMimic中使用的7520-14.3型号。

PD设定点（目标位置）通过以下关系计算得出：

$$\boldsymbol{q}^{\mathrm{tar}} = \boldsymbol{q}^{0} + \boldsymbol{\alpha} \odot \mathbf{a}$$

其中 $\boldsymbol{q}^{0}$ 表示默认关节位置，a是策略输出的动作。预裁剪控制力矩随后通过标准PD控制律计算：

$$\boldsymbol{\tau} = k_p(\boldsymbol{q}^{\mathrm{tar}} - \boldsymbol{q}) - k_d\dot{\boldsymbol{q}}$$

在此表达式中，q和 $\dot{q}$分别表示当前关节位置和速度。

H. 评估协议和指标

本小节详细介绍了论文中使用的评估协议和指标。我们的目标是评估在符合训练条件且能反映仿真到真实性能的控制环境下，跟踪保真度、鲁棒性和泛化能力。

a) 运动分割和片段长度：为了避免因运动时长不同引入的偏差，所有参考运动都被分割成固定长度的10秒片段（对应500个控制步），这与训练期间使用的片段长度一致。时长不足10秒的运动无需进一步分割进行评估。模拟和真实世界中的所有报告指标均在片段级别进行计算。

b) 仿真评估设置：在仿真评估过程中，我们保留了训练期间使用的相同传感器噪声和基础域随机化，包括质心偏移和默认关节位置偏移。这确保了评估反映基本的仿真到现实鲁棒性，而不是理想化的无噪声跟踪。每个运动片段在10次独立的运行中评估，并且这些试验的指标被平均。

理想化的无噪声跟踪。每个运动片段在10次独立的运行中评估，并且这些试验的指标被平均。

c) 未见运动集：除了从训练分布中抽取的测试运动外，我们还评估了在未见运动集上的泛化能力。该集合通过从重定向的AMASS CMU和KIT动捕序列中均匀采样1000个运动片段（每个10秒）构建而成，明确排除了所有用于训练或选为极端运动的运动。未见集合主要包含行进、转向和简单的舞蹈类行为，用于评估超越精选极端运动库的泛化能力。

d) 评估指标：我们采用一组在拟人机器人运动跟踪中常用的基于姿态和物理感知的指标。所有指标均按片段计算，然后对片段结果进行平均。

设 $t = 1, \ldots, T$ 为一个回合中控制步的索引，步长为 Δt （秒），设 $i = 1, \ldots, N$ 为被追踪的躯干。所有与躯干相关的误差都是基于相同的被追踪躯干子集计算的。

e) 根相对躯干位置：参考运动通过躯干姿态与机器人对齐。我们记对齐后的参考躯干位置为 $\mathrm{p}_t^{\mathrm{ref}},_i$，机器人躯干位置为 $\mathrm{p}_t^{\mathrm{rob}}{}_i$，.

f) MPJPE (mm).:

$$\mathrm{MPJPE} = 1000 \cdot \frac{1}{T}\sum_{t=1}^{T}\left(\frac{1}{N}\sum_{i=1}^{N}\left\|\mathbf{p}_{t,i}^{\mathrm{ref}} - \mathbf{p}_{t,i}^{\mathrm{rob}}\right\|_2\right). \quad (14)$$

g) Δvel (mm/帧)：设 $\mathrm{v}_t^{\mathrm{ref}},_i$ 和 $\mathrm{v}_t^{\mathrm{rob}},_i$ 为线性速度（单位：m/s）。

$$\Delta v = 1000 \cdot \Delta t \cdot \frac{1}{T}\sum_{t=1}^{T}\left(\frac{1}{N}\sum_{i=1}^{N}\left\|\mathbf{v}_{t,i}^{\mathrm{ref}} - \mathbf{v}_{t,i}^{\mathrm{rob}}\right\|_2\right). \quad (15)$$

h) Δacc (mm/帧^2)：加速度通过速度的有限差分计算：

$$\mathbf{a}_{t,i} = \frac{\mathbf{v}_{t,i} - \mathbf{v}_{t-1,i}}{\Delta t}.$$

每步的加速度误差为：

$$e_t^{\mathrm{acc}} = \frac{1}{N}\sum_{i=1}^{N}\left\|\mathbf{a}_{t,i}^{\mathrm{ref}} - \mathbf{a}_{t,i}^{\mathrm{rob}}\right\|_2. \quad (16)$$

我们排除环境重置后的立即步骤。报告的加速度误差为：

$$\Delta a = 1000 \cdot \Delta t^2 \cdot \frac{1}{T}\sum_{t=1}^{T} e_t^{\mathrm{acc}}. \quad (17)$$

i) 成功率：如果一个场景因超时（即运行了其全部分配的时长）而结束，而不是提前终止，则被视为成功。评估期间使用的终止标准与训练期间使用的标准相同，包括违反安全阈值。成功率定义为成功场景的比例。

TABLE XI: **Skill-level grouping of real-world evaluation motions.** Each skill category groups multiple motion instances with similar semantic meaning and dynamic structure. Motion IDs correspond to retargeted motions used during hardware evaluation.

Skill category	Motion IDs
Flip	5, 9, 10, 15, 16, 18, 22
Handspring	2, 3, 4, 19, 23
Acrobatics	1, 6, 7, 20, 21
Breakdance	11, 12, 13, 14
Martial arts	8, 17, 38

j) Aggregation.: MPJPE, Δv, and Δa are first averaged over control steps within each episode. Final reported metrics are obtained by averaging over all evaluated episodes (and over motion clips when evaluating multiple motions).

I. Skill-level Grouping for Real-world Evaluation

For real-world evaluation, we organize individual motions into a set of *extreme skill categories*. Each skill groups multiple motion instances that share similar semantic intent and high-level dynamic structure. This grouping is used *only for reporting and analysis*, and does not affect training or policy execution.

The classification is based on human semantic understanding of the motions, taking into account characteristic body coordination patterns, contact configurations, and dominant dynamic features (e.g., aerial phases, rapid contact switching, or impulsive landing). We emphasize that this grouping is not obtained through automated clustering or learned representations, but serves as a transparent and interpretable abstraction for summarizing real-world performance.

Tab. XI lists the correspondence between skill categories and the underlying motion instances used in hardware evaluation.

J. Motion Subsets Used in Fidelity–Scalability Analysis

In the fidelity–scalability analysis Sec. IV-D, we evaluate how tracking performance degrades as the size of the motion library increases. To ensure a controlled and interpretable comparison, we explicitly specify the motion subsets used at each scale.

We begin with a curated set of extreme motions that exhibit high angular velocities, rapid contact switching, and challenging balance conditions. These motions represent a diverse yet highly demanding subset of the full motion library. All quantitative evaluations in Q2 are conducted on the same **first 10 motions** of this set, which include fast flips, contact-rich acrobatics, and extreme balance behaviors. By fixing the evaluation set, we isolate the effect of training-scale growth from changes in evaluation difficulty.

To study scalability, we progressively expand the training motion library to include 10, 20, and all extreme motions. The additional motions are drawn from the same extreme motion pool and increase diversity in contact patterns and dynamic regimes, while preserving a consistent difficulty level. This

TABLE XII: **Motion subsets used in Q2 fidelity–scalability analysis.** All evaluations are performed on the same first 10 extreme motions. Larger training sets (20 and all motions) extend this core set with additional diverse extreme motions.

Training motions	Motion IDs
10	3–10, 13, 14
20	2–10, 13-22
50	All

design yields a controlled yet representative benchmark for analyzing the fidelity–scalability trade-off under increasingly diverse extreme motions. Tab. XII summarizes the motion IDs used at each scale.

K. Knee negative power penalty.

For each knee joint j with torque τ_j and joint velocity $\dot{q}_j$, we define the instantaneous mechanical power $P_j = \tau_j \dot{q}_j$ and penalize excessive *negative* power (large braking torques) beyond a deadband threshold. Concretely, for knee joints selected via the regex `".*_knee_joint"` we compute

$$\tilde{P}_j = \max\big(0,\ -P_j - 150\big), \tag{18}$$

where the deadband is 150 (in the same units as power, e.g., W). The normalized per-step penalty is

$$c_{\text{knee}} = \sum_{j \in \mathcal{J}_{\text{knee}}} \left(\frac{\tilde{P}_j}{500} \right)^2, \tag{19}$$

where 500 is the normalization constant `power_norm`. In the reward function this term is added with weight $w = -10$, i.e., $r_{\text{knee}} = w\, c_{\text{knee}}$, so that large negative joint power at the knees is strongly discouraged.

L. TensorRT-Accelerated Deployment

For deployment on the real G1 robot we run the model as an ONNX graph, executed by ONNX Runtime with TensorRT acceleration. At run time the deploy node continuously receives joint states and IMU signals from the robot, builds a compact observation vector, and feeds it to the ONNX base policy. The base policy outputs a 29-dimensional whole-body action, and, the residual policy produces a small corrective action in the same space; the two are added to obtain the final command. This command is interpreted as desired joint positions, which are tracked by a PD controller with safety envelopes on torques and velocities, and converted into low-level motor commands at a control period of $20\,\text{ms}$.

When a GPU with TensorRT is available, ONNX Runtime delegates the forward pass of these ONNX models to TensorRT, giving a lower-latency implementation of the same controller. In our implementation, we use **5 ODE integration steps** (often informally referred to as denoising steps), which strike a favorable balance between action quality and computational cost. All inference is executed on the onboard **NVIDIA Jetson Orin NX** of the Unitree G1 humanoid robot. With TensorRT acceleration, the end-to-end policy inference latency, including the full ODE integration procedure, is approximately

表 XI：真实世界评估动作的技能等级分组。每个技能类别将具有相似语义意义和动态结构的多个动作实例分组。动作 ID 对应硬件评估期间使用的重定向动作。

技能类别	动作 ID
Flip	5, 9, 10, 15, 16, 18, 22
手翻	2, 3, 4, 19, 23
杂技	1, 6, 7, 20, 21
霹雳舞	11, 12, 13, 14
武术	8, 17, 38

j) 汇总：MPJPE、 Δv和 Δa首先在每个场景内的控制步骤中进行平均。最终报告的指标通过对所有评估场景（在评估多个动作时还包括对动作片段）进行平均获得。

I. 技能等级分组用于真实世界评估

为了进行真实世界评估，我们将单个动作组织成一组极端技能类别。每个技能类别包含多个具有相似语义意图和高级动态结构的动作实例。这种分组仅用于报告和分析，不会影响训练或策略执行。

该分类基于人类对动作的语义理解，考虑了特征的身体协调模式、接触配置和主导的动态特征（例如，空中阶段、快速接触切换或冲动式落地）。我们强调这种分组不是通过自动聚类或学习表示获得的，而是作为一种透明且可解释的抽象方法，用于总结真实世界的表现。

表XI列出了技能类别与硬件评估中使用的底层运动实例之间的对应关系。

J. 用于保真度—可扩展性分析的Motion子集

在保真度—可扩展性分析第IV-D节中，我们评估了随着运动库规模的增加，跟踪性能如何下降。为了确保受控且可解释的比较，我们明确指定了每个尺度上使用的运动子集。

我们从一组经过精心挑选的极端运动开始，这些运动表现出高角速度、快速接触切换和具有挑战性的平衡条件。这些运动代表了完整运动库中一个多样化且要求极高的子集。Q2中的所有定量评估都在该集合的前10个运动上进行的，这些运动包括快速翻转、富含接触的杂技和极端平衡行为。通过固定评估集，我们隔离了训练尺度增长的影响，以及评估难度变化的影响。

为研究可扩展性，我们逐步扩展训练运动库，使其包含10个、20个以及所有极端运动。新增的运动来自相同的极端运动库，增加了接触模式和动态状态多样性，同时保持了一致的难度水平。这

表XII：用于Q2保真度—可扩展性分析的Motion子集。所有评估都在同一组前10个极端运动上进行的。更大的训练集（20个和所有运动）通过添加更多多样化的极端运动扩展了这个核心集合。

训练动作	动作ID
10	3–10, 13, 14
20	2–10, 13-22
50	All

这一设计提供了一个可控且具有代表性的基准，用于分析在日益多样化的极端运动下保真度与可扩展性之间的权衡。表XII总结了每个尺度下使用的运动ID。

K. 膝关节负功率惩罚。

对于每个膝关节 j ，具有扭矩τ_j 和关节速度$\dot{q}_j$，我们定义瞬时机械功率 $P_j = \tau_j \dot{q}_j$，并惩罚超出死区阈值的过度负功率（大制动力矩）。具体来说，对于通过正则表达式 ".*_knee_joint" 选择的膝关节，我们计算

$$\tilde{P}_j = \max\big(0,\ -P_j - 150\big), \tag{18}$$

死区为150（与功率单位相同，例如W）。每步的归一化惩罚为

$$c_{\text{knee}} = \sum_{j \in \mathcal{J}_{\text{knee}}} \left(\frac{\tilde{P}_j}{500} \right)^2, \tag{19}$$

其中500是归一化常量功率_norm。在奖励函数中，此项以权重 $w = -10$（即$r_{\text{knee}} = w\, c_{\text{knee}}$）添加，以强烈避免膝盖处的大幅负关节功率。

L. TensorRT加速部署

在真实G1机器人上部署时，我们以ONNX图的形式运行模型，由ONNX Runtime配合TensorRT加速执行。运行时，部署节点持续从机器人接收关节状态和IMU信号，构建紧凑的观测向量，并将其输入ONNX基础策略。基础策略输出29维全身动作，残差策略在同一空间内产生小幅修正动作；两者相加得到最终指令。该指令被解释为期望的关节位置，由具有扭矩和速度安全边界的PD控制器跟踪，并在20ms控制周期内转换为低级电机指令。

当可用支持 TensorRT 的 GPU 时，ONNX 运行时将这些 ONNX 模型的前向传递委托给 TensorRT，从而为相同的控制器提供更低延迟的实现。在我们的实现中，我们使用 5 个 ODE 积分步（通常非正式地称为去噪步），在动作质量和计算成本之间取得了良好的平衡。所有推理都在 Unitree G1 仿人机器人的板载 NVIDIA Jetson Orin NX 上执行。通过 TensorRT 加速，包括完整的 ODE 积分过程在内的端到端策略推理延迟，大约为每控制步 10ms。

10 ms per control step. This latency comfortably satisfies real-time control requirements and enables closed-loop execution of high-dynamic whole-body motions entirely on the onboard compute, without reliance on offboard processing.

M. Failure Cases and Discussion

While OMNIXTREME achieves strong sim-to-real performance overall, we observe a small number of failure cases during real-world deployment. These failures predominantly occur during highly impulsive landing phases of certain extreme motions, where large transient braking loads trigger hardware protection mechanisms, including motor overcurrent, power limits, or battery undervoltage events.

Notably, these motions can be executed reliably in simulation and sim-to-sim transfer, indicating that failures are not caused by degraded tracking accuracy or loss of balance. Instead, they expose residual discrepancies between simulated actuation models and the true hardware capability envelope under extreme dynamic conditions. These observations suggest that further improving robustness for extreme motions will require more comprehensive modeling of real actuator and power-system limits, including the coupled effects of torque, speed, current, power flow, and battery voltage dynamics, which remain challenging to capture accurately in simulation.

In the current work, post-training is performed via a lightweight residual policy that refines a frozen flow-based base controller. While this design offers stability and sample efficiency, it may also limit the extent to which the full representational capacity of the large flow-based model can be further adapted to hardware-specific constraints. Future research may explore more native post-training strategies [51] that directly fine-tune or adapt the full base policy under actuation-aware objectives. Such approaches could potentially unlock greater expressivity and hardware alignment from large generative control models, enabling more principled integration between flow-based policy learning and real-world physical constraints.

这种延迟足以满足实时控制要求，并能够完全在板载计算上执行高动态全身运动的闭环控制，而无需依赖外部处理。

M. 失败案例与讨论

尽管 OMNIXTREME 在整体上实现了强大的仿真到现实的性能，但在实际部署过程中，我们观察到少量失败案例。这些失败主要发生在某些极端动作的高度冲击着陆阶段，其中大的瞬态制动力触发了硬件保护机制，包括电机过流、功率限制或电池欠压事件。

值得注意的是，这些动作在仿真和仿真到仿真的迁移中可以可靠地执行，这表明失败并非由跟踪精度下降或失去平衡引起。相反，它们暴露了在极端动态条件下，模拟驱动模型与真实硬件能力边界之间的残余差异。这些观察表明，为了进一步提高极端动作的鲁棒性，需要更全面地建模真实执行器和电源系统的限制，包括扭矩、速度、电流、功率流和电池电压动态的耦合效应，而这些效应在仿真中仍然难以准确捕捉。

在当前工作中，通过一种轻量级残差策略进行后训练，该策略细化了一个冻结的基于流的基控制器。虽然这种设计提供了稳定性和样本效率，但也可能限制大型基于流模型的全表示能力在多大程度上可以进一步适应特定的硬件约束。未来的研究可以探索更原生的后训练策略 [51]，在驱动感知目标下直接微调或适应完整基策略。这类方法有可能从大型生成控制模型中解锁更大的表达性和硬件对齐能力，实现基于流的策略学习与现实世界物理约束之间更原则性的集成。